MODULE 5A: OEM SPECIFIC OVERVIEW

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GENERAL MOTORS

Learning Objectives: GM

1. ***GM first to implement EDR***
2. ***Six different generations, all <u>very</u> different***
3. ***Industry leading data limitations specific to each EDR family***
4. ***Clear rules for recording and overwriting old records***

Learning Objectives - GM

5. ***Too complicated to memorize it all - Just Read the Data Limitations***
6. ***Multiple Modules with data in them – SDM (ACM), ROS, ASCM, FCM, and OCS – with future releases expected for additional legacy FCM's.***

6 groups of GM SDM Recorders

1. **94-01 300ms of X Delta V only**
2. **99-12 150 ms of X Delta V only AND**

   5 samples PRE-CRASH @ 1 sec intervals

3. **04-12 300 ms of X and Y Delta V AND**

   5 samples PRE-CRASH @ 1 or 0.5 sec intervals

"3.5" 10-12 holds 3 events, does not erase ND's

4. **12-now Part 563 intent – 300 ms X-Y**

   10 samples pre crash @0.5 sec intervals

5. **SDM50 in 2020+ Corvette**

   16 samples pre crash @0.5 sec intervals

   More data elements including stability control

GM EDR Evolution

GEN 1 94-01	GEN 2 99-2012	GEN 3.0 05-2012	GEN 3.5 2010-2012 454 Cable	GEN 4 2012+	SDM50 2020+
No Precrash	Precrash 5 @ 1 sec for BOTH events	Precrash 5 @ 0.5 or 1 sec ONE SET of precrash	Precrash 5 @ 0.5 or 1 sec For each event	Precrash 10 @ 0.5 sec	Precrash 10 or 16 @ 0.5 sec + T_O
2 EVENTS - FIRST D, One ND/ DLE, replace SDM after D			3 Events, numbered ND's or D's, replace after 3D's		+event summary 30 records +2 PedPro +3 AACN
Record on frontal airbag algo wakeup		Record on 5mph X or Y DV			
300ms XDV =wakeup, must be neg	150ms XDV, must be neg, 50ms before D 100 after	300ms X+Y DV, + or - some D's -70 to +220, some 300ms after wakeup		300ms after wakeup	120G *accelerometer*

GM EDR Evolution #2

GEN 1 94-01	GEN 2 99-12	GEN 3.0 05-12	GEN 3.5 10-12 454 Cable	GEN 4 2012+	SDM50 2020+
Erase ND after 250 Key Overwrite old ND if bigger 80%, overwrite with newest event 20% RTFDL			Never Erase. Only overwrite ND's if >250KC old		
			Integrated ROS		
		Separate ROS '07	Accel Data 0-300ms		
			Concurrent event flag		Add strg, Brake pres., 4 wheel spds, Time of Crash. SC Long & Lat accel +Yaw in 2021+

Chevrolet SDM Phase In 21_5 87%

	GREEN = 300 ms Longitudinal Delta V					Red = Precrash Data & 150ms Longitudinal Delta V						Blue = Precrash Data & 300ms XY Delta V					3 Events		Part 563 Intent								SDM50	GM		
Model	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Astro			2888	2888	2888	2888	2829	2829	2829	2829	2829	2829																		
Avalanche									2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	*									
Aveo																3320	3320	3320		454	*				454					
Beat (India)																					454	454								
Blazer					2888	2888	2829	2829	2829	2829	2829	2829*	*													454	454	454	454	454
Bolt EV																									454	454	454	454	454	454
Camaro			2888	2888	2888	2829	2829	2829	2829								454	454	454	454	454	454	454	454	454	454	454	454	454	454
Caprice	3002	3002	3002															452	454	police	police	police	police	police	police					
Captiva																	454	454	454	454	*	Sport	Sport							
City Express																						780	780	780	780	780				
Colorado											3003	3003	3003	3003	3003	3003	3003	3003	3003			454	454	454	454	454	454	454	454	852
Corvette				2888	2888	2829	2829	2829	2829	2829	2829	3003	3003	3003	3003	3003	3003	3003	3003	3003	454	454	454	454	454	454	845	845	845	845
Cruze																	454	454	454	454	454	454	454	454	454	454	454			
Equinox												3004	3004	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454	454	454
Express			2888	2888	2888	2888	2888	2829	2829	3003	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454
Impala		3002	3002				2829	2829	2829	2829	2829	2829	3320	3320	3320	3320	3320	3320	454	454	454	454	454	454	454	454	454			
Kodiak										2829	2829	2829	2829	2829	2829	2829			2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Low Cab Fwd																										454				
Malibu				2888	2888	2888	2829	2829	2829	2829	3321***	3321 mix	3321	3321	3321	3321	3321	3321	3321	3321 (2012	454	454	454	454	454	454	454	454	454	454
Matiz																			454											
Orlando																			454	454	454									
Silverado				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS*mix	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454	454	852
Sonic																			454	454	454	454	454	454	454	454	454			
Spark	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011		454	454	454	454	454	454	454	454	454	454	
SSR										3003	3003	3003	3003									SS 454	454	454						
Suburban				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	845	845	852
Tahoe				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	845	845	852
Tracker			DLC	DLC	DLC	DLC	DLC	DLC	DLC	DLC	DLC														454	454	454	454	454	
TrailBlazer									2829	2829	2829	3293	3293	3293	3293	3293												454	454	454
Traverse																3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454	454	454
Trax																				454	454	454	454	454	454	454	454	454	454	
Volt																		454	454	454	454	454	454	454	454	454				
	14%	48%	76%	94%	100%	20%	74%	86%	96%	98%	100%	14%	43%	71%	79%	82%	91%	92%	22%	96%		100%		100%						
PHASE IN	Delta V	Delta V	Delta V	Delta V	Delta V	Pre-Crash	Pre-crash	Pre-crash	Pre-Crash	Pre-Crash	Pre-Crash	Pre + XY	Pre + XY	Pre + XY	Pre + XY	Pre + XY	Pre + XY	Pre + XY	563	563	563	563								

X DV

+ SPEED

+Y DV

563

3 events

SDM50

GM chart printed in [illegible] and is on student reference [illegible]

Freeze Frame Data

** = 150ms of Longitudinal Delta V instead of 300ms

563

Chevrolet Recent Changes

1 1 1 animation

	Part 563 Intent								SDM50	GM				
Model	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Astro														
Avalanche	454	454	*											
Aveo		454	*				454							
Beat (India)			454	454										
Blazer								454	454	454	454	454	454	454, EV 852
Bolt EV							454	454	454	454	454	454	454	
Camaro	454	454	454	454	454	454	454	454	454	454	454	454	454	
Caprice	**454**	police	police	police	police	police	police							
Captiva	454	454	*	**Sport**	**Sport**									
City Express				**780**	**780**	**780**	**780**	**780**						
Colorado	3003			**454**	**454**	**454**	**454**	**454**	454	454	454	**852**	**852**	**852**
Corvette	3003	3003	454	454	454	454	454	**454**	**845**	**845**	**845**	**845**	**845**	**845**
Cruze	454	454	454	454	454	454	454	**454**	**454**					
Equinox	454	454	454	454	454	454	454	**454**	**454**	454	454	454	454	**852**
Express	3320 ROS	3320 ROS	3320 ROS	3320 ROS	454	454	454	**454**	**454**	454	454	454	454	454
Impala	**454**	454	454	454	454	454	454	**454**	**454**					
Kodiak	**2012**	**2013**	**2014**	**2015**	**2016**	2017	2018	2019	2020	2021	2022	2023	2024	2025
Low Cab Fwd								454						
Malibu	3321	3321 (2012	454	454	454	454	454	454	**454**	454	454	454	454	
Matiz	454													
Orlando	454	454	454											
Silverado	454	454	454	454	454	454	454	454	**454**	**454**	**454**	**852**	**852**	**852**
Silverado Med Duty												**454**	454	454
Sonic	454	454	454	454	454	454	454	454	**454**					
Spark		454	454	454	454	454	454	454	**454**	**454**	**454**			
SSR				**SS**	**454**	**454**	**454**							
Suburban	454	454	454	454	454	454	454	454	**454**	**852**	**852**	**852**	**852**	
Tahoe	454	454	454	454	454	454	454	454	**454**	**852**	**852**	**852**	**852**	
Tracker							454	454	**454**	**454**	**454**			
TrailBlazer										**454**	**454**	**454**	**454**	**454**
Traverse	454	454	454	**454**	**454**	**454**	**454**	454	**454**	**454**	**454**	**454**	**454**	**852**
Trax		454	454	454	454	454	454	454	**454**	**454**	**454**		**454**	**454**
Volt	454	454	454	454	454	454	454	454						12

No Precrash in 2013-5 Express/Savanna

SDM 50 phasing in SLOWLY

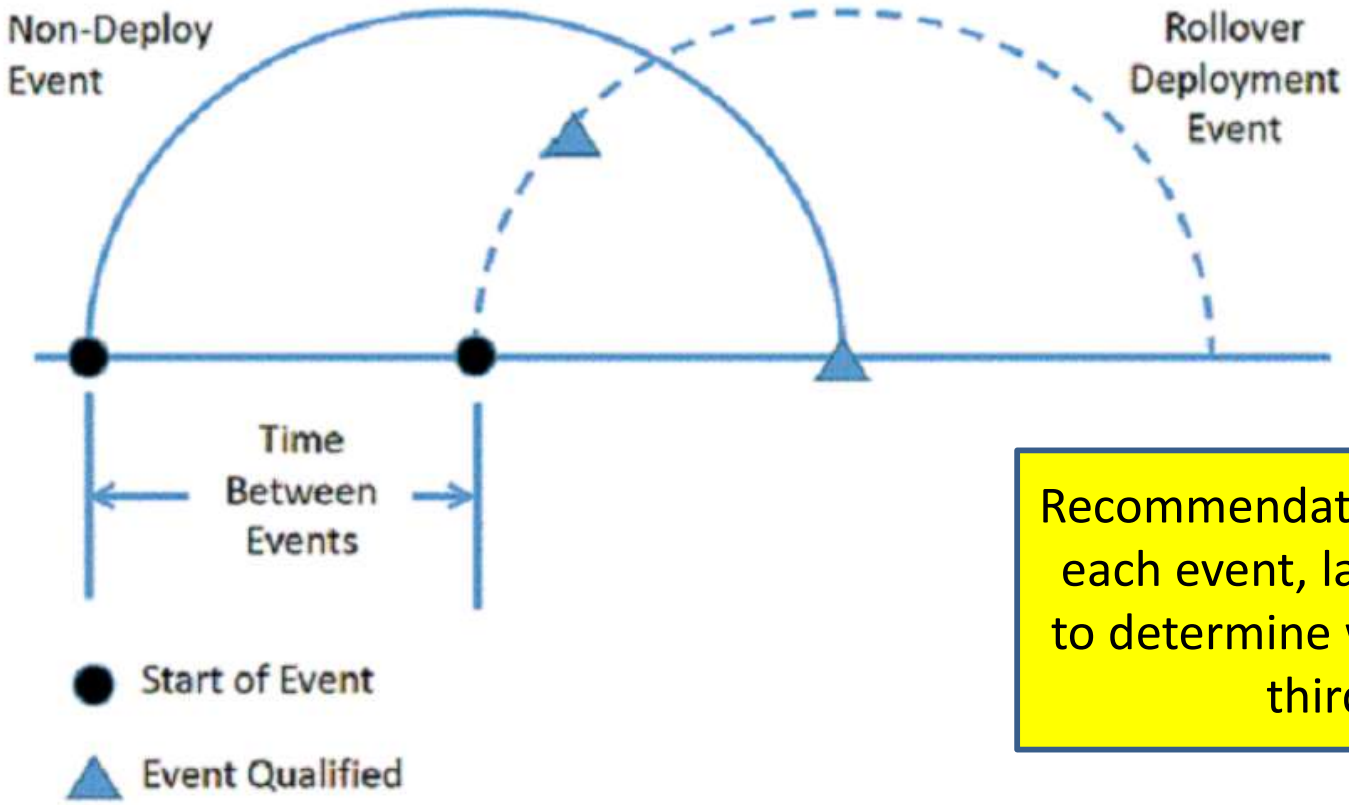
New for 2010+ – Concurrent Event Flag

- Indicates that two algorithms are running at the same time.
- Time between events is from start of algorithm of one to start of algorithm of the other
- Order of events in memory is by when they ***qualified*** as recordable, **not** by AE time
- An event can start later but qualify sooner and be first in memory

See graphic added to data lims next slide

2010+ Concurrent Event Flag DL

Example of a Concurrent Event:
A Non-Deployment event begins. Before the Non-Deployment event is qualified, a Rollover Deployment event begins and is qualified. Sometime after the Rollover event is qualified, the Non-Deployment event is qualified. The Rollover event will be recorded in the first open record even though the Non-Deployment event enabled before the Rollover event. The Non-Deployment event will be recorded in the next open record. The "Concurrent Event Flag Set" parameter will indicate "Yes" for the Non-Deployment event. The "Time Between Events" parameter will indicate the time from the start of the Non-Deployment event to the start of the Rollover event.



Recommendation: Print out speed data from each event, lay side by side, line up speeds to determine which event was first, second, third…. Update graphic

Event Record #1	Event Record #2
Event record Type = Rollover	Non-deployment
Concurrent Event Flag = No	Concurrent Event Flag = Yes
Time Between Events = N/A	Time Between Events = XX seconds

When Concurrent Event Flag is Yes, Overlay Precrash Data from all events

- The event number gives when the event QUALIFIED for recording, not when I started
- Time between events is time between consecutive wake-ups, NOT always time between qualifying events
- Recap Delta V or, if rollover, event type over precrash data to assist understanding

Record 1 -57X, 39Y Event 1 of 5 Time Between Events NA	Record 2 Rollover Event 2 of 5 Time Between Events 0.75	Record 3 +1X, -5Y Event 4 of 5 TBEv 1.01
	47 [75]	**Speed, Vehicle Indicated (Vehicle Speed) (MPH [km/h])**
45 [73]	45 [73]	
44 [71]	44 [71]	
43 [69]	43 [69]	
41 [66]	41 [66]	41 [66]
38 [61]	38 [61]	38 [61]
31 [50]	31 [50]	31 [50]
30 [49]	30 [49]	30 [49]
30 [49]	30 [49]	30 [49]
23 [37]	23 [37]	23 [37]
11 [17]		11 [17]
		17 [27]
		Data Not Available
		Data Not Available

Gen 5 SDM50 2020+ **8 sec.** Precrash Data

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 1 of 6

Time (sec)	Accelerator Pedal, % Full (%)	ABS Activity	Brake Pedal Position (%)	Driver Applied Brake Pedal Pressure (kPa)	Driver Applied Brake Pedal Pressure Detected	Brake Pedal Moderate Travel Achieved
-8.0	0	Off	0	0	False	False

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 2 of 6

Time (sec)	Cruise and Speed Limiter Switch Status	Cruise Secondary Switch Status	Cruise Control Active	Adaptive Cruise Control Active Authenticated	Transmission Estimated Gear	Transmission Shift Lever Position	Reduced Engine Power Mode Indicator On
-8.0	No Activation	No Activation	Inactive	Inactive	Ninth	Forward Range A	Off

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 3 of 6

Time (sec)	Engine RPM (RPM)	Engine Torque (foot-pounds)	Engine Throttle, % Full (%)	Speed, Vehicle Indicated (MPH [km/h])	Yaw Rate in 2021	Wheel Angular Velocity (RPM)	Wheel Angular Velocity (RPM)
-8.0	1,301	50.5	16	44.1 [71]		595	598

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 4 of 6

Time (sec)	Right Rear Wheel Angular Velocity (RPM)	Left Rear Wheel Angular Velocity (RPM)	Steering Wheel Angle (deg)	Propulsion System Active	Run/Start Input	Ignition Prolongation Time (sec)	Secondary Collision Prolongation Timer (sec)
-8.0	599	598	0	True	High	5.0	0.0

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 5 of 6

Time (sec)	Antilock Brake System Failed	Brake Pedal Override Flag	Traction Control System Present	Traction Control System Failed	Traction Control System Enabled	Traction Control System Active	Traction Control System Driver Intent
-8.0	False	False	True	False	True	Traction Control Off	Enabled

IMU data for 2020 now deleted, was bad before 21.0

SDM-50 Data Validated

- SAE 2025-01-8714

Analysis of the Event Data Recorder vehicle system data of a GM Vehicle Intelligence Platform Airbag Control Module (SDM50)

Brian Smyth, Charles L Crosby, Ryan Bickhaus, James Smith, Exponent Inc; Dustin Edmunds, Donald Floyd, Vipul Modi, RaShawndra D. Outlaw, Jeff Wright, General Motors LLC

GM **A**ctive **S**afety **C**ontrol **M**odule

- A separate module located in the trunk of the car with data on automatic emergency braking and other collision avoidance technologies
- Before 2019 only downloadable by GM, released in CDR 17.9 to get raw data file via DLC only for future interpretation. Now we get interpretation and have direct to module cables.
- REQUIRES CDR 900 interface!
- Application chart follows

GM ASCM CDR 23.4 Coverage

BUICK	ASCM Location	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
ENCLAVE	cargo left front of wheel well						833 1B						
ENVISION	rear seat left trim by wheel wl				833 1A	833 1B				854 3-LC			
LACROSSE	trunk left					833 1B							
REGAL	trunk left						833 1B						
CADILLAC		**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
ATS	trunk left	833 1A					833 1B						
CT4	Trunk upper shelf								854 3-LC	855 3-HC			
CT5	Trunk upper shelf								854 3-LC	855 3-HC			
CT6	Primary right, secondary left				833 1A		833 1B, 834 2						
CTS SEDAN	trunk left		833 1A				833 1B						
CTS	trunk left				833 1A		833 1B						
ESCALADE	under left qtr window trim			833 1A			833 1B			854/855 LC/HC			
LYRIC	cargo left front of wheel well											833 1B	
SRX	under left rear seat	833 1A											
XT4	cargo left front of wheel well							833 1B					
XT5	cargo left front of wheel well					833 1B							
XT6	trunk left								833 1B		833 1B/834 2		
XTS	trunk left	833 1A					833 1B						
CHEVROLET		**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
BOLT	trunk left										834 2		
BLAZER	trunk left							833 1B					
MALIBU	trunk left				833 1A	833 1B							
SILVERADO	rear wall behind middle seat											855	
SUBURBAN	under left qtr window trim										854 3-LC		
TAHOE	under left qtr window trim										854 3-LC		
TRAVERSE	trunk right						833 1B						
VOLT	trunk left					833 1B							
GMC		**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
ACADIA	cargo left front of wheel well					833 1B							
HUMMER EV											855 3-HC		
SIERRA	rear wall behind middle seat											854 3-LC	
YUKON	under left qtr window trim									854 3-LC		854/855	

GM ASCM Evolution

EOCM 1A 2013+ 833 Cable	EOCM 1B 2016+ 833 Cable	EOCM 2 2018+ SUPERCRUISE 834 Cable	EOCM3 – Lo 2020+ Cable 854	EOCM 3 – Hi 2021+ Cable 855
1 Event			3 events	
Forward Collision Braking Events			Deployment and Nondeployment Events, Ped Warning, AEB events	
Data -3.92 to 0, at 80ms intervals			Data -4.9 TO +3, OR - 7.9 TO 0 for Deploys @0.1 or 0.5 sec intervals	

GM Active Safety Control Module Help File Callout

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4	2020 Cadillac	CT5 (note 6)	ACM	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 845	Center tunnel
			ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
4	2020 Cadillac	CT6 (note 6)	ASCM (Note2) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 834	In the luggage compartment, right center, mounted to the right side in the battery well. The secondary ASCM is on the left.
			ASCM (Note3)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, right center, mounted to the right side

Supercruise UKL

w/o Supercruise UKN

Two different ASCM's for UGN and UKL

GM Active Safety Module



- Must be downloaded separately from ACM
- Icon will appear after VIN input for any vehicle that MIGHT the optional ASCM module
- Module /data will only be present if option “UGN” or “UKL” (Super Cruise) has been purchased for the subject vehicle.

GM Active Safety Module

- 1A/1B - Data can be captured every 80 milliseconds and can record up to 50 records for a total of 4 seconds of data. The data recording window can be either 4 seconds before the end of an auto braking event or 4 seconds before an airbag deployment. ONE EVENT ONLY
- Data is recorded in non-volatile memory at the next proper power-down of the module. Data may **not** be stored if the battery power is lost before the power-down.

Active Safety Control Module Summary of Some Key Data

- Odometer
- Collision imminent
- GPS Position
- Driver Override
- Master clock
- Steering Wheel Angle
- Braking system auto brake status
- Accel Actual position
- Vehicle Speed average non driven wheels
- Commanded Transmission Gear
- Target relative velocity (Closing Speed!)

Screenshots from ASCM

1G6KT5R66KU***** Page 4 of 32 Printed on: Friday, September 27 2019 at 15:48:31

Vehicle Action Data, -3.92 to 0.00 sec - Table 1 of 2

Time (sec)	Master Clock (Sec)	Event Type	Odometer (miles [km])	GPS Latitudinal Position (degrees)	GPS Longitudinal Position (degrees)	Vehicle Speed (MPH [km/h])	Yaw Rate (radians/sec)
-3.92	576,102	Collision Imminent Braking	1893 [3046]	42.5848	-83.6947	6 [10]	0.00
-3.84	576,182	Collision Imminent Braking	1893 [3046]	42.5848	-83.6947	6 [10]	0.00

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Vehicle Action Data, -3.92 to 0.00 sec - Table 2 of 2

Time (sec)	Vehicle Lateral Acceleration (m/s^2)	Vehicle Longitudinal Acceleration (m/s^2)	Commanded Transmission Gear
-3.92	0.188	-0.500	Fourth
-3.84	0.156	-0.375	Fourth
-3.76	0.125	-0.375	Fourth

Screenshots from ASCM

1G6KT5R66KU****** Page 8 of 32 Printed on: Friday, September 27 2019 at 15:48:31

Driver Selection and Action Data, -3.92 to 0.00 sec - Table 1 of 4

Time (sec)	Brake Pedal Position (%)	Brake Pedal Initial Travel Achieved	Driver Initiated Braking	Accelerator Actual Position (%)	FAB Current Setting Value	Driver Selected Cruise Type	Adaptive Cruise Control Cruise Mode
-3.92	0.0	False	False	0.4	Alert and Brake	Adaptive Cruise	Standby Enabled
-3.84	0.0	False	False	0.8	Alert and Brake	Adaptive Cruise	Standby Enabled
-3.76	0.0	False	False	0.8	Alert and Brake	Adaptive Cruise	Standby Enabled

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Driver Selection and Action Data, -3.92 to 0.00 sec - Table 2 of 4

Time (sec)	Adaptive Cruise Control Active	Adaptive Cruise Control Disengage Reason	Lane Centering Control State	Lane Centering Control Enabled	Lane Centering Control Disengage Reason	Lane Centering Control Driver Override	FAB Driver Override Status - Pre-Automatic Braking Accelerator Override
-3.92	False	Brake Pressed	Disabled Flash Red	False	2C	False	False
-3.84	False	Brake Pressed	Disabled Flash Red	False	2C	False	False

Screenshots from ASCM

1G6KT5R66KU****** Page 13 of 32 Printed on: Friday, September 27 2019 at 15:48:31

Driver Selection and Action Data, -3.92 to 0.00 sec - Table 3 of 4

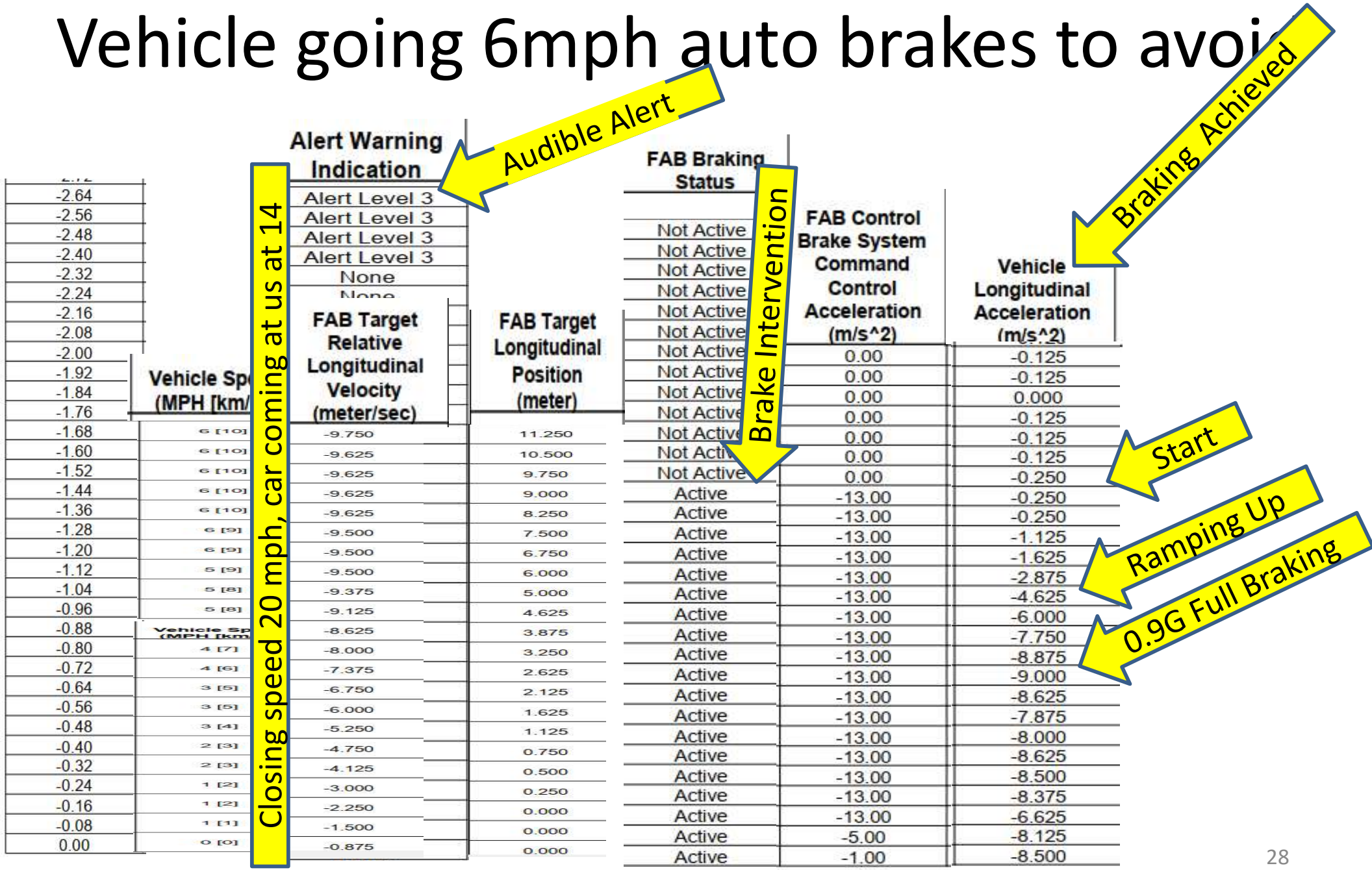
Time (sec)	FAB Driver Override Status - Steering Wheel Motion Inhibit	FAB Driver Override Status - Post-Autonomous Braking Accelerator Pedal Override	FAB Driver Override Status - Driver Braking	Driver Attention State	Driver Hands on Steering Wheel	Steering Wheel Touch Zone 1	Steering Wheel Touch Zone 2
-3.92	False	False	True	On Road	True	True	True
-3.84	False	False	True	On Road	True	True	True
-3.76	False	False	True	On Road	True	True	True

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Driver Selection and Action Data, -3.92 to 0.00 sec - Table 4 of 4

Time (sec)	Steering Wheel Touch Zone 3	Vehicle Ahead Indication	Alert Warning Indication
-3.92	True	Alert Level 1	None
-3.84	True	Alert Level 1	None
-3.76	True	Alert Level 1	None
-3.68	True	Alert Level 1	None
-3.60	True	Alert Level 1	None
-3.52	True	Alert Level 1	None
-3.44	True	Alert Level 1	None
-3.36	True	Alert Level 1	None
-3.28	True	Alert Level 1	Alert Level 3
-3.20	True	Alert Level 1	Alert Level 3
-3.12	True	Alert Level 1	Alert Level 3

Selected data -braking to avoid collision
Vehicle going 6mph auto brakes to avoi

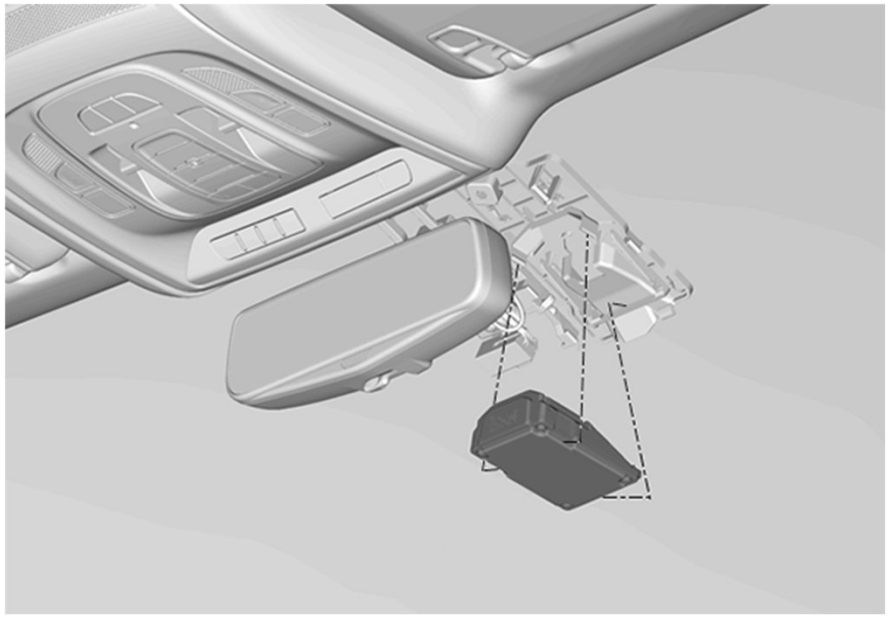


	Vehicle Sp (MPH [km/	FAB Target Relative Longitudinal Velocity (meter/sec)	FAB Target Longitudinal Position (meter)	Alert Warning Indication	FAB Braking Status	FAB Control Brake System Command Control Acceleration (m/s^2)	Vehicle Longitudinal Acceleration (m/s^2)
-2.64				Alert Level 3	Not Active		
-2.56				Alert Level 3	Not Active		
-2.48				Alert Level 3	Not Active		
-2.40				Alert Level 3	Not Active		
-2.32				None	Not Active		
-2.24				None	Not Active		
-2.16					Not Active		
-2.08					Not Active		
-2.00					Not Active	0.00	-0.125
-1.92					Not Active	0.00	-0.125
-1.84					Not Active	0.00	0.000
-1.76					Not Active	0.00	-0.125
-1.68	6 [10]	-9.750	11.250		Not Active	0.00	-0.125
-1.60	6 [10]	-9.625	10.500		Not Active	0.00	-0.125
-1.52	6 [10]	-9.625	9.750		Not Active	0.00	-0.250
-1.44	6 [10]	-9.625	9.000		Active	-13.00	-0.250
-1.36	6 [10]	-9.625	8.250		Active	-13.00	-0.250
-1.28	6 [9]	-9.500	7.500		Active	-13.00	-1.125
-1.20	6 [9]	-9.500	6.750		Active	-13.00	-1.625
-1.12	5 [9]	-9.500	6.000		Active	-13.00	-2.875
-1.04	5 [8]	-9.375	5.000		Active	-13.00	-4.625
-0.96	5 [8]	-9.125	4.625		Active	-13.00	-6.000
-0.88	Vehicle Sp (MPH [km	-8.625	3.875		Active	-13.00	-7.750
-0.80	4 [7]	-8.000	3.250		Active	-13.00	-8.875
-0.72	4 [6]	-7.375	2.625		Active	-13.00	-9.000
-0.64	3 [5]	-6.750	2.125		Active	-13.00	-8.625
-0.56	3 [5]	-6.000	1.625		Active	-13.00	-7.875
-0.48	3 [4]	-5.250	1.125		Active	-13.00	-8.000
-0.40	2 [3]	-4.750	0.750		Active	-13.00	-8.625
-0.32	2 [3]	-4.125	0.500		Active	-13.00	-8.500
-0.24	1 [2]	-3.000	0.250		Active	-13.00	-8.375
-0.16	1 [2]	-2.250	0.000		Active	-13.00	-6.625
-0.08	1 [1]	-1.500	0.000		Active	-5.00	-8.125
0.00	0 [0]	-0.875	0.000		Active	-1.00	-8.500

ASCM Validation SAE 2024-01-2888

- **Analysis of the Event Data Recorder (EDR) Function of a GM Active Safety Control Module (EOCM3 LC)**
- A series of vehicle Automatic Emergency Braking (AEB) test maneuvers were conducted, triggering the ASCM Event Data Recorder (EDR) function. The vehicle dynamic state was independently monitored and recorded by on-board instrumentation and compared to the ASCM recorded data.

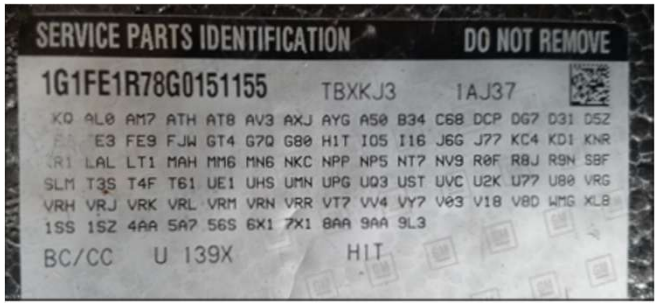
Front Camera Module (FCM)



FCMs are mounted on the windshield near the rear-view mirror.

GM FCM EDR

- GM released the ability to read data from Front Camera Modules(FCM) with V 21.2 of the CDR Software (Date: 8/2021)
- The data consists of photos taken before, during, and after the crash.
- Not all GM cars are equipped with FCMs
- Some cars have FCMs but are not supported at this time
- We can identify which cars have an FCM by
  - Looking for presence of the camera module near the rear-view mirror
  - Verifying the (RPO) UEU and UGN in the vehicles Service Parts Identification sticker, or through the QR code on the FMVSS Sticker (see photos)





Combined ASCM and FCM Access 23.4

Model	Model Year											
3 digits Bosch Cl	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
BUICK												
ENCLAVE						ASCM						
ENCORE								FCM				
ENVISION				ASCM			FCM					
LACROSSE					ASCM							
REGAL						ASCM						
CADILLAC	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
ATS							FCM					
CELESTIQUE												ASCM+FCM
CT4								ASCM				
CT5								ASCM				
CT6 w/o UKL					ASCM	without supercruise						
CT6 w/UKL				834		Supercruise Option Code UKL						
CTS SEDAN		ASCM										
CTS				ASCM			FCM					
ESCALADE			ASCM									
LYRIC											ASCM	
SRX	ASCM											
XT4							ASCM					
XT5					ASCM							
XT6									ASCM			
XTS							FCM					
CHEVROLET	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
BLAZER							ASCM					
BOLT											ASCM/FCM	
CAMARO											FCM	
CRUZE							FCM					
EQUINOX							FCM				FCM	
MALIBU				ASCM								
SILVERADO							FCM					
SPARK							FCM					
SUBURBAN											ASCM	
TAHOE											ASCM	
TRAVERSE						ASCM						
VOLT					ASCM		FCM					
GMC	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
ACADIA					ASCM							
HUMMER EV											ASCM	
SIERRA							FCM				ASCM/FCM	
SIERRA HD											FCM	
TERRAIN							FCM					
YUKON									ASCM			

GM FCM EDR

- The Lookup file will identify which GM vehicles may have an FCM.

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4,6	2019 Cadillac	XTS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel, forward
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, under the left side trim panel
			FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield

ACM Data may be present

ASCM Data may be present

FCM Data may be present

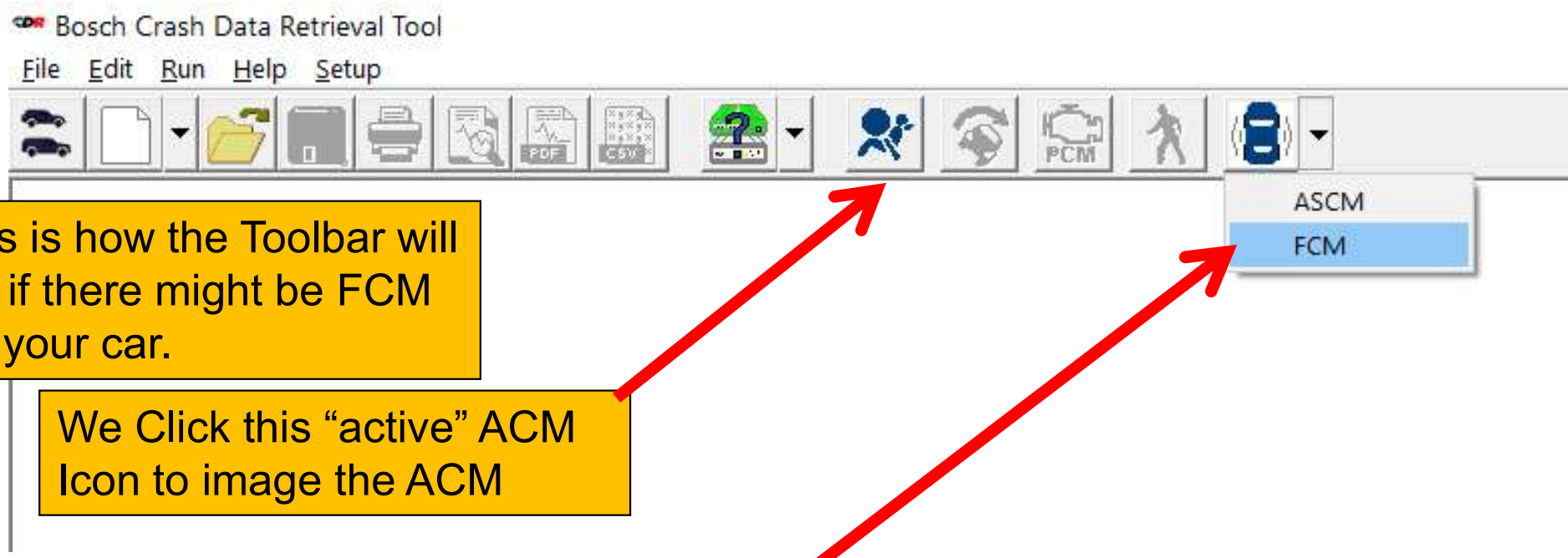
GM FCM EDR

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4,6	2019 Cadillac	XTS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel, forward
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, under the left side trim panel
			FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield

Note we get hints on how to Download the FCM – How to image

GM ASCM EDR – Bosch CDR 900



And this is how the Toolbar will appear if there might be FCM data in your car.

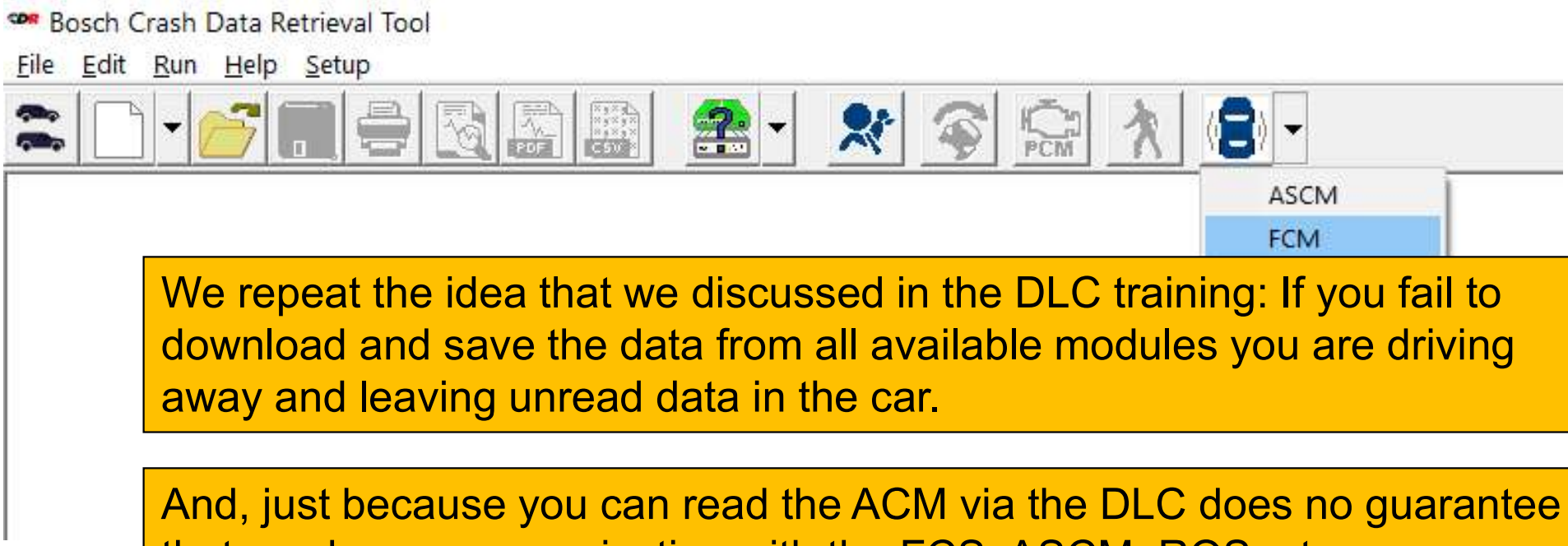
We Click this “active” ACM Icon to image the ACM

Then once the ACM Download is complete and saved, we click this “active” Dropdown Icon to image the FCM and then save that file as well

The FCM may take 20-30 minutes to download – be sure your jump pack is charged up at the start!

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GM ASCM EDR – Bosch CDR 900



We repeat the idea that we discussed in the DLC training: If you fail to download and save the data from all available modules you are driving away and leaving unread data in the car.

And, just because you can read the ACM via the DLC does no guarantee that you have communication with the FCS, ASCM, ROS, etc .

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GM FCM EDR

- There are several network pathways in modern vehicles
- Damage may allow DLC communication to one module and prevent comm to another module.
- Do the research to see if your car is equipped with all indicated modules (ACM, ASCM, FCM) and read as necessary (DLC or DTM or maybe one of each)
- At this time FCMs are only available on 2019MY and later GM's
- If you previously read the SDM and ASCM on an FCM now-supported vehicle, **you may have to return to read the data from this new module type**
- The FCM raw data was NOT saved in any ASCM files you may have read.

GM FCM EDR

- When a CDR file contains image data, the user can export the data using options from the CDR software:

CDR Tool icon



We Click this “Icon to export FCM images

GM FCM EDR

- The Front Camera Module (FCM) can record two events. Recording is triggered by “events of interest”. The following are possible “events of interest” that trigger data recording listed in the **priority order**:
- **Airbag Deployment Event,**
- **Pretensioner Deployment Event,**
- **Non-Deployment Event,**
- Pedestrian Braking,
- Collision Imminent Breaking (CIB)
- Pedestrian Warning,
- Front Collision Alert,
- Lane Departure Warning.

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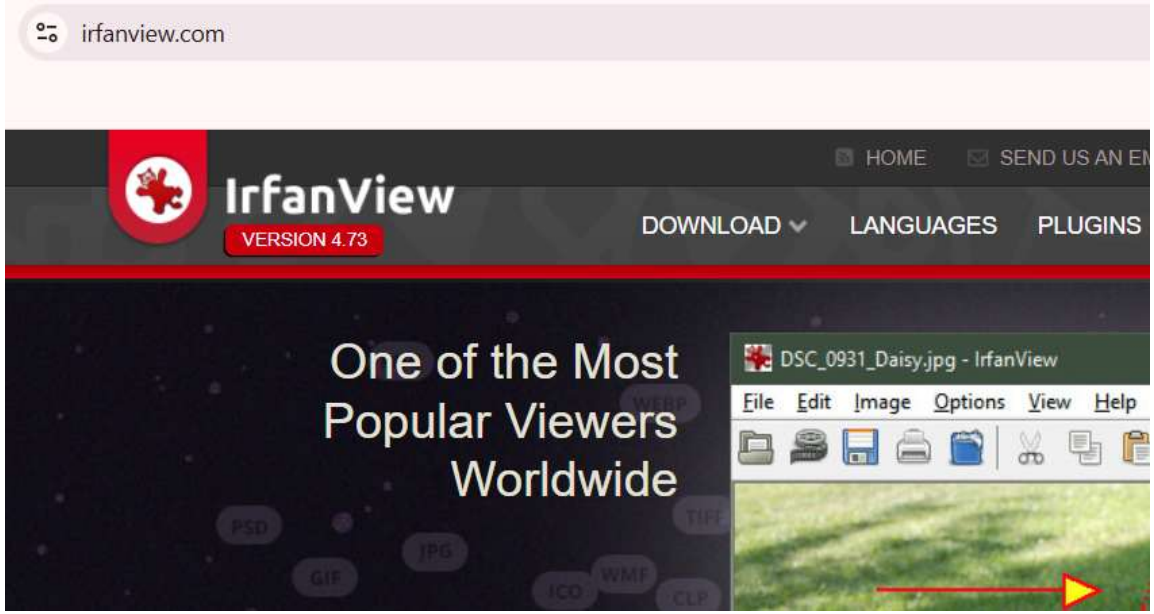
GM FCM EDR

- The new event is stored only if it is a higher priority than the priority of one of the currently stored events. The new event overwrites the lowest priority event that is already stored.
- If all previously stored events are of equal priority (and are lower priority than the new event), the oldest stored event is the one being overwritten.
- The image data is not rendered directly in the CDR program.
- It must be exported to a file in the images' native format.

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GM FCM EDR

- Photos are black and white *.pgm files (**P**ortable **G**ray **M**ap)
- You will need a *.pgm file viewer – many free *.pgm apps are available.
  (Try IrfranView (https://www.irfanview.com/)
- Users can regenerate the images collected by the FCM as many times as they wish recreating the file as it existed when the data was collected from the vehicle.
- Yet again we see that having the Raw file IS the evidence – As an analyst, *if sharing with another,* you want the raw FCM cdrx file, not someone else's downloaded *.pgm images.
- The data in the FCM CDRx file remains untouched and cannot be changed.
- The images can be downloaded and generated from the Raw file as many times as necessary.



GM FCM EDR

- If the exported image file appears to be changed by another user, you can re-export the image directly from the CDR file it was created; therefore, removing any doubt that the data source was changed after the file was generated.
- One more issue… The FCM image will be upside down.



GM FCM EDR

- The solution is to “Invert” or Flip the photo. DO NOT “rotate” it.

- Original FCM “upside down” photo



- A successfully “inverted” or “flipped” photo
  (if you convert pgm to jpg, grab the photo at the top and drag the top thru the bottom of the photo to invert it)





- A confusing, inaccurate “rotated” photo.

(Note car appears to be driving on *wrong side of road*)

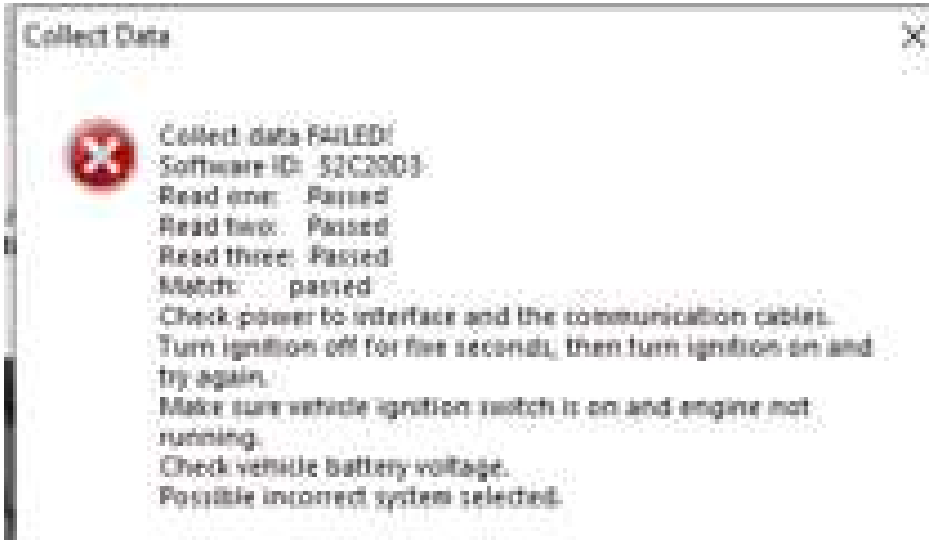


43

Issues obtaining FCM data

JAN 2025 POST inspected a 24 Equinox. The vehicle was involved in an severe offset frontal collision. When we arrived, the vehicle was covered with plastic wrap. The wrap was removed and we found the FCM plug hanging in the hole where the windshield was. The plug appeared intact and undamaged. as we entered the car, we found the FCM along with the mirror on the console. The FCM appears in good shape. We attempted a benchtop of the FCM several times with software version 24.2.383. The software would run 3 passes and then attempt to run pass 1 of the image. It would then hang up and restart the 3 passes then we would get the screen below. NO communications and turn ignition off for 5 seconds. We attempted about 4 times getting the same error.

We brought the module back to the office and today we rolled back to software to Version 23.3. We have done several FCMs with 23.2 including a 24 Equinox. We rolled back the CDR900 and attempted to image the FCM again several times and got the same runs below



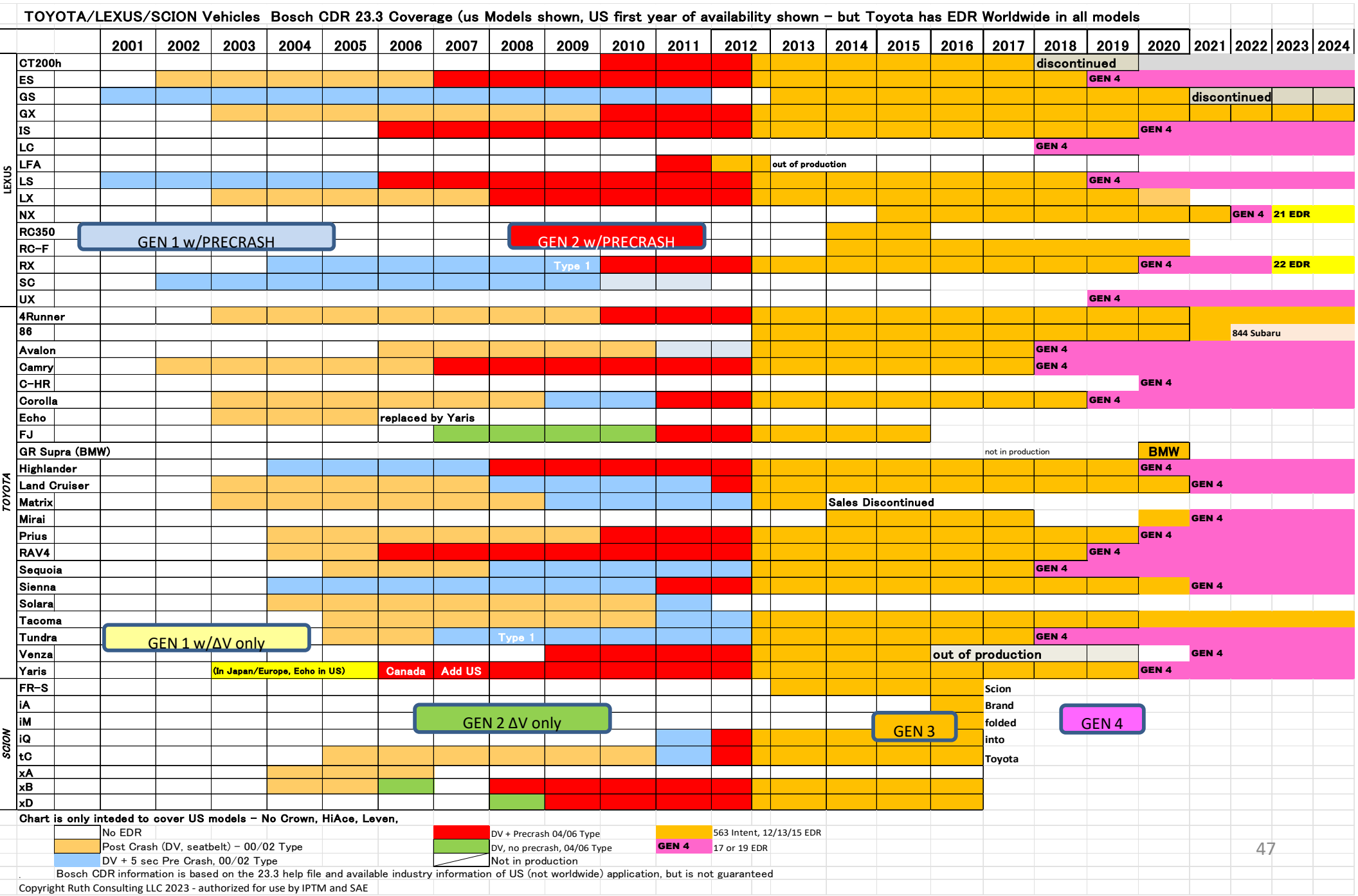
4 different modules to read in GM Occupant Classification Sensor

Mkt	Year/Make	Model	Module	New CDR2		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4	2024 Chevrolet	Silverado EV	ACM	N/A	Data	CDR 900	CDR 900 - click here	Cable ID # 852 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note2)	N/A		CDR 900	CDR 900 - click here	CDR 900 - Cable# 870	Below rear driver's side seat.
			FCM how to image	N/A	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield
ar Buttons			OCS	CDR 2		CDR 900	CDR 900 - click here	D2M downloads are not supported	Under Passenger Seat

OCS will provide detail on the front passenger classification (empty, child, small adult, etc)

TOYOTA

Toyota CDR 23.3 Coverage US Models



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Toyota Unique Help File Organization

Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in so
note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Click here for a list of s

Click here for tips on how to use this *Supported Vehicles* help topic

IMPORTANT: To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the ap

General Toyota Application Notes:

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will n
vehicles listed in the help file below are supported as indicated by the list and any associated coverage notes.

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287)

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vel

Vehicle Specific Application Notes:

Click here to view all vehicle specific Toyota coverage notes

Vehicles Are Listed Alphabetically
Not by Model Year

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W Y Z

4Runner (Toyota) **2003 and newer** **Market: 7**

Year	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
2002 - 2008	ACM	Data - 02EDR	CANplus or CDR 900	CANplus - F00K108287 - *for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to*	CANplus - (no adapter) Cable#F00K108617 CDR 900 - click here (use listed cable above and no ACM adapter)	Under Center Stack

New Toyota Vehicle Lookup

Camry (Toyota) **2001 and newer** **Market: 7**

7=All Markets the Vehicle is Sold

Year	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
2001 - 2004 (all markets) 2005 (US/Canada)	ACM	Data - 00EDR / 02EDR	CANplus or CDR 900	CANplus - F00K108287 *- for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to vehicles sold outside the US &* CDR 900 - click here	CANplus - (no adapter) Cable#F00K108615 - (no adapter) Cable#F00K108617 Click here [?] to match the ACM part number to the correct cable CDR 900 - click here (use listed cable above and no ACM adapter)	Under Center Stack

Multiple Cables for same model

Must Look up Cable for DTM by Part No

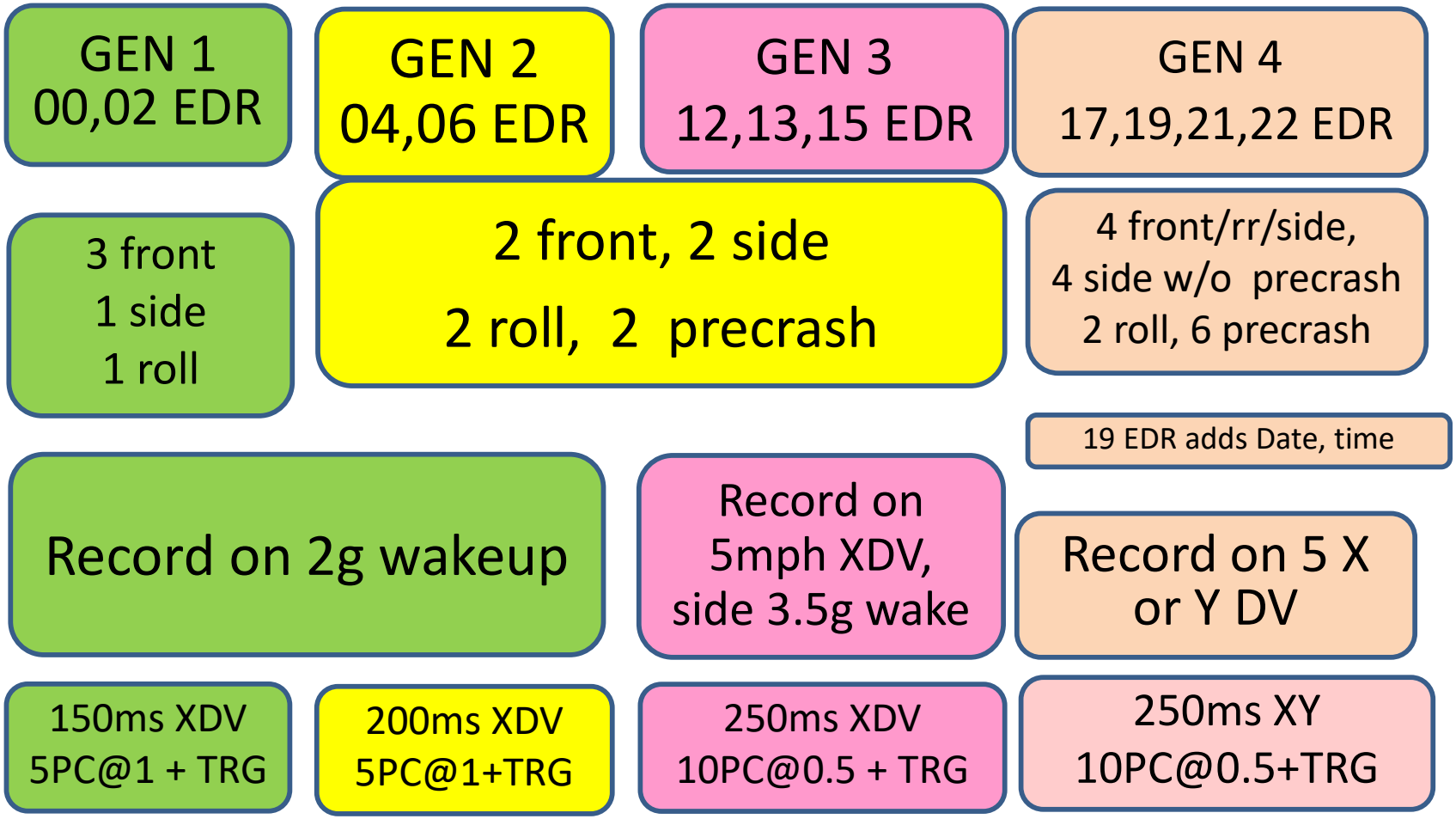
New Toyota Help File ***U.S. Issues***

- Because the help file applies WORLDWIDE, it lists MANY models not found in the US/CAN.
- Toyotas internal records may be by ***CALENDAR*** YEAR, not ***MODEL*** YEAR. Help File may show EDR is first available in 2003, but that was 2004 Model Year in the US. MOST models are “one year off”.
- Avalon shows EDR in 02 but prior charts showed no EDR until 06 – do not assume EDR present in every vehicle listed in the help file

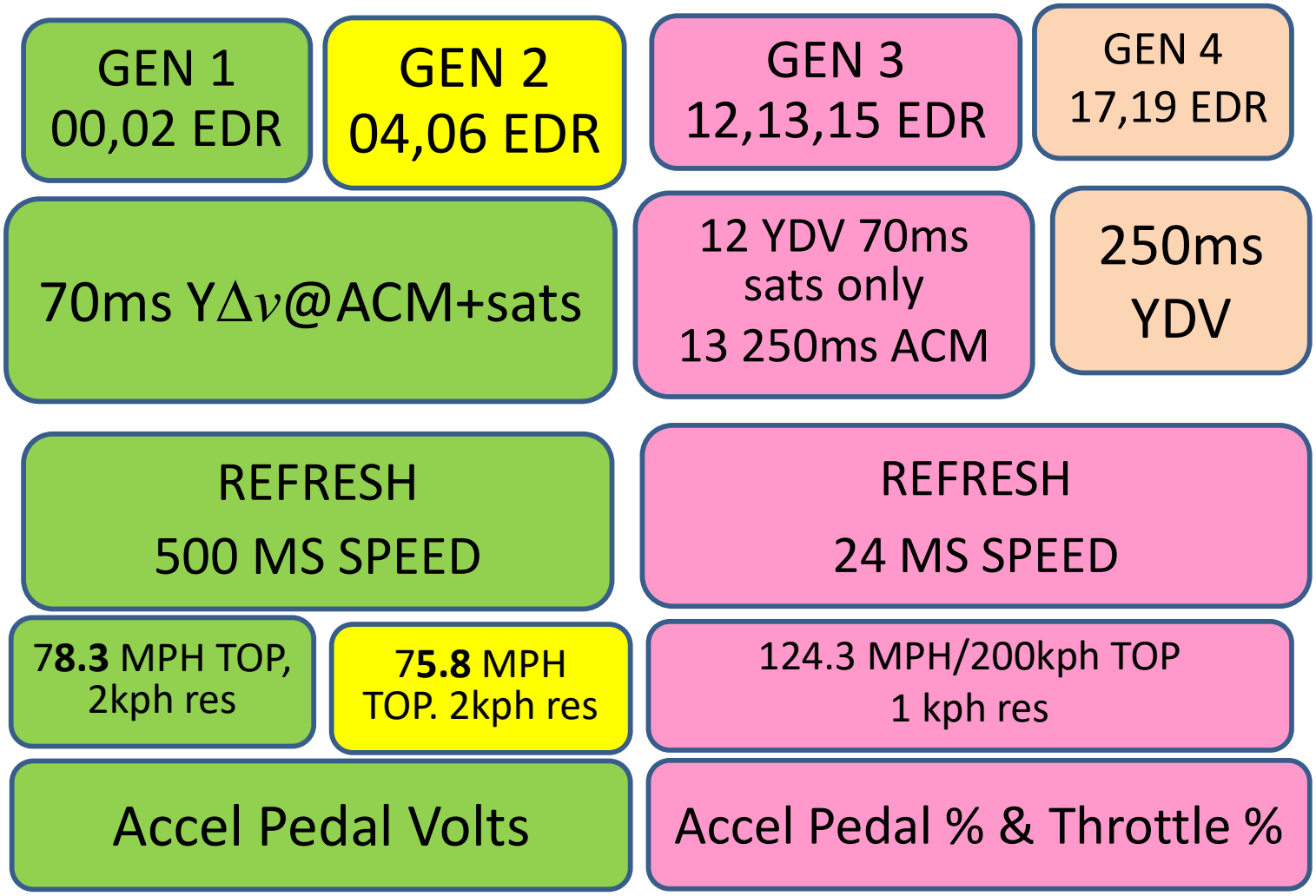
Unique to Toyota

- Delta V is recorded real time as it is occurring (GM waits until event is over to write anything). Even in older models with no backup power supply for recording, partially written Delta V is possible
- There are multiple “recording complete” entries:
- 1. Complete file
- 2. Crash Pulse subsection (Delta V), per event
- 3. Precrash subsection (Speed), per event

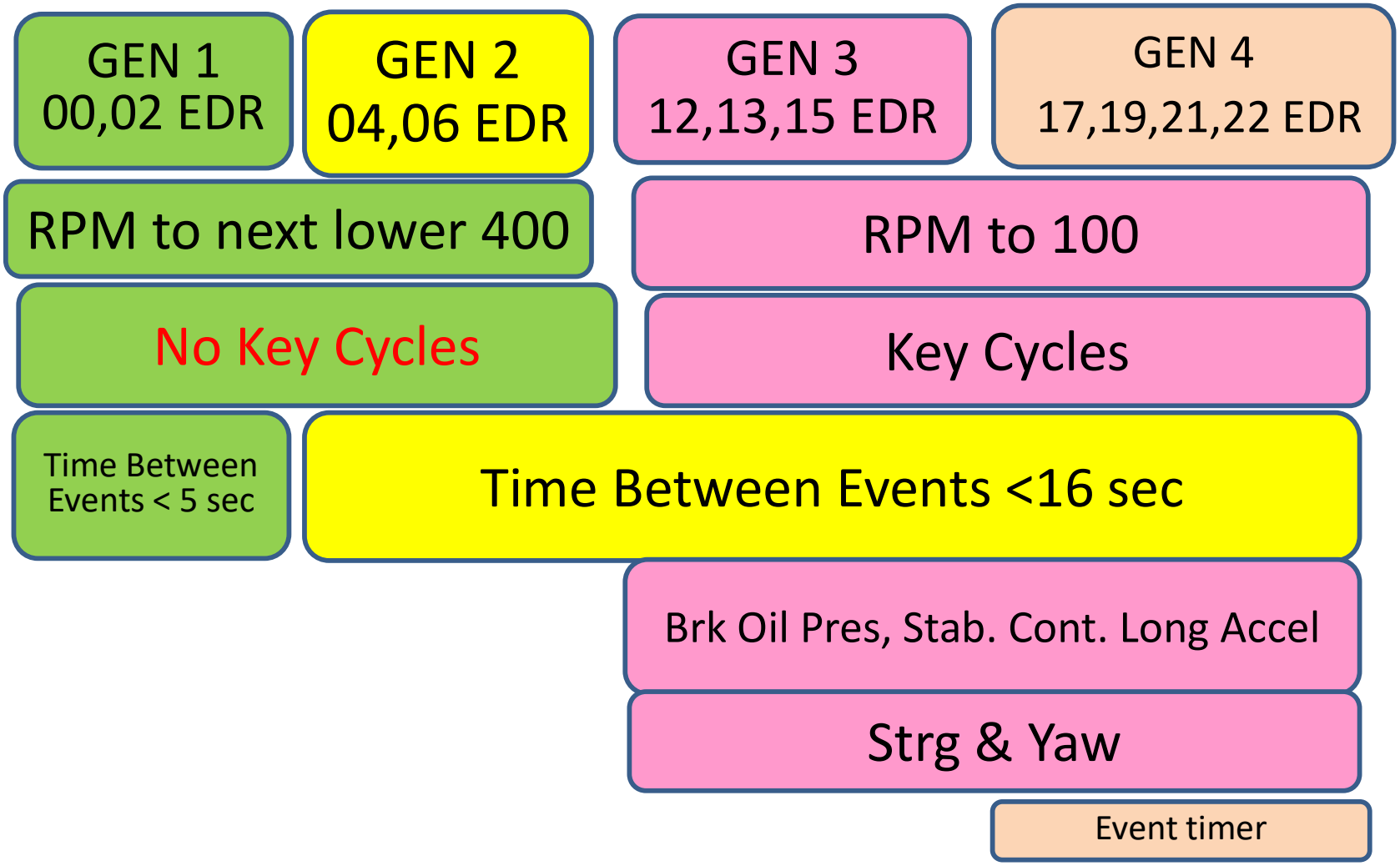
Toyota EDR Evolution



Toyota EDR Evolution cont'd



Toyota EDR Evolution cont'd



00/02 EDR record summary

Front/Rear Event Record Summary at Retrieval

Side Events not numbered

Events Recorded	TRG Count	Crash Type	Time (msec)	Event & Crash Pulse Data Recording Status
Most Recent Frontal/Rear Event	4	Front/Rear Crash	0	Complete (Front/Rear Page 1)
1st Prior Frontal/Rear Event	3	Front/Rear Crash	-5120 or greater	Complete (Front/Rear Page 2)
Prior Frontal/Rear Event	1	Front/Rear Crash	N/A	Complete (Front/Rear Page 0)

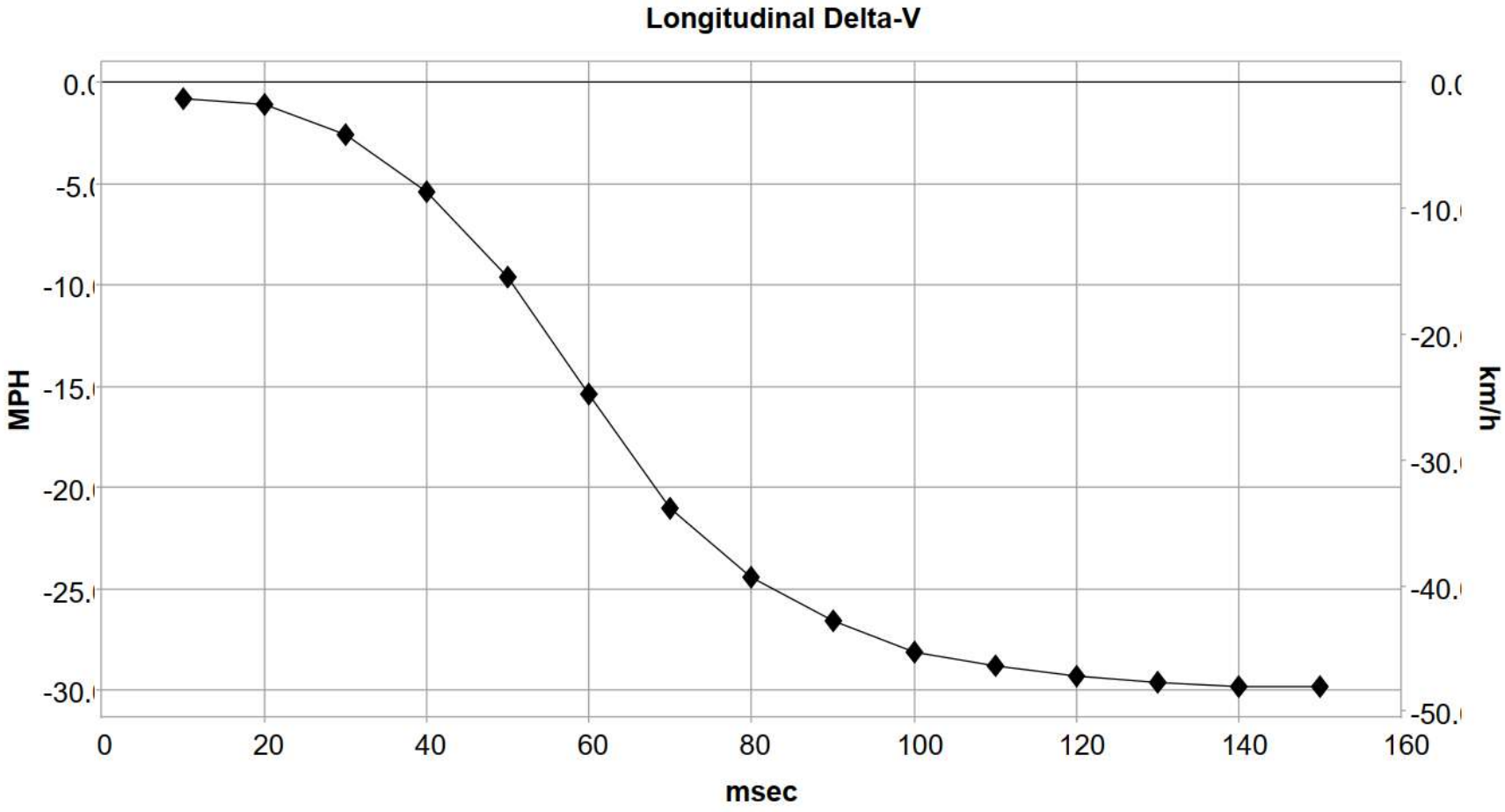
- Saves **third** front/rear event that came closest to deploying airbags (most negative DV).
- 1st prior event #3 = -5120ms ***or greater*** before most recent event, means ***NOT RELATED***
- TRG 2 was less important than TRG 1 and was overwritten when TRG 4 came in.

Gen 1 (and 2) Max Long. Δv

- "Max" on top is the biggest number in the data table. Due to shorter duration recording, this is usually also very near the end of the crash.

Longitudinal Crash Pulse (Most Recent Frontal/Rear Event, TRG 2 - table 1 of 2)

Max Longitudinal Delta-V (MPH [km/h])	-29.9 [-48.1]



Finding PDOF in Toyota Gen1&2

- Toyota Gen 1&2 record frontal data when frontal algorithm wakes up, side data **only** when side algorithm wakes up and side data is supported.
- Many early Toyotas do **not** support side recording, you must estimate PDOF from visible crush damage.
- In later Toyotas with side recording, look for “pairs” of front and side recordings. Side recording duration is about 70ms, get PDOF by taking Y value at 70ms and X value from 70ms. BE CAREFUL – Side algorithm may wake up at a different time than frontal, **try to synchronize them.**

Unique to Toyota

- Toyota records ADDITIONAL precrash data point at TRG and the **time** from *last regular interval* recording to TRG (resolution to 0.1 sec, 0.0 to 0.9, truncated down)
- Last data may be WRITTEN at TRG, but could still be "old" as speed data in **Gen 1 & 2** was only transmitted every 0.5 sec. If speed data did not change from prior data point, it may be old by 0.5 sec. If the last reported value is different, it cannot be older than the time from last value to prior value.

Gen 1 & 2 Pre-Crash Data

- Accel Pedal Data in ***Voltage from sensor***
- Zero accel pedal is typically 0.78V, 100% is 3.79 V max seen to date but data limitations indicates the low and high could be different.
- You can calculate your own percentage if you want. %=(V-0.78)/(3.79-0.78) Update .5 sec
- RPM in 400 RPM increments, 400-799=400, 800-1199=800, etc. Updates every 24ms.
- Brake on/off updates immediately, no delay (accel pedal may as on when brake is on, but they are from different times)

Unique Toyota Memory Structure

- Toyota records pre-crash in a different memory section than Delta V. It gives records "page numbers" 0 and 1 and tells you which Precrash page goes with which Delta V page.
- If there are two or more events within 0.5 sec, all may be linked to the same precrash page. This is noted in the event record summary. CDR will print the SAME precrash page 4 times
- Precrash Page 0 is stored at TRG4, not TRG7

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	7	Side Crash	0	Complete (Page 0)	Complete (Side Page 0)
1st Prior Event	6	Front/Rear Crash	-20	Complete (Page 0)	Complete (Front/Rear Page 1)
2nd Prior Event	5	Side Crash	-228	Complete (Page 0)	Complete (Side Page 1)
3rd Prior Event	4	Front/Rear Crash	-282	Complete (Page 0)	Complete (Front/Rear Page 0)

Un-linked precrash data

- The remaining precrash space is then known as “UNLINKED PRE CRASH” and was likely linked to a previous event. It is printed after the last linked precrash page
- In multiple event scenarios, Unlinked precrash data overlaps linked precrash and shows driver behavior farther back in time. Always check for overlap!

System Status at Time of Retrieval

ECU Part Number	89170-52D80
ECU Generation	06EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretensioner Deployment
Contains Unlinked Pre-Crash Data	Yes

Overlay Unlinked Data with Linked

Pre-Crash Data, -5 to 0 seconds (1st Prior Event, TRG 8)

Time (sec)	-4.1	-3.1	-2.1	-1.1	-0.1	0 (TRG)
Vehicle Speed (MPH [km/h])	74.6 [120]	70.8 [114]	70.8 [114]	69.6 [112]	36 [58]	36 [58]
Brake Switch	ON	OFF	OFF	OFF	OFF	OFF
Accelerator Rate (V)	0.78	0.78	1.05	1.45	0.78	0.78
Engine RPM (RPM)	2,400	2,400	3,200	3,200	1,200	800

Pre-Crash Data, -5 to 0 seconds (Unlinked, Page 1)

Time (sec)	-4.7	-3.7	-2.7	-1.7	-0.7	0 (TRG)
Vehicle Speed (MPH [km/h])	75.8 [122]	75.8 [122]	74.6 [120]	70.8 [114]	70.8 [114]	69.6 [112]
Brake Switch	ON	ON	ON	OFF	OFF	OFF
Accelerator Rate (V)	0.78	0.78	0.78	0.78	1.05	1.56
Engine RPM (RPM)	3,600	2,800	2,400	2,400	3,200	3,200

These two precrash data sets are from the same crash sequence

Toyota Gen 2 Side Crash Report

BOSCH

CDR CRASH DATA RETRIEVAL SYSTEM

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	4T1BF3EK4BU664098
User	Richard R. Ruth, P.E.
Case Number	Gazdick salvage yard find
EDR Data Imaging Date	06/01/2011
Crash Date	
Filename	4T1BF3EK4BU664098_ACM_GAZDICK_BODY_SHOP.CDRX
Saved on	Wednesday, June 1 2011 at 21:52:48
Collected with CDR version	Crash Data Retrieval Tool 4.0
Reported with CDR version	Crash Data Retrieval Tool 4.0
EDR Device Type	airbag control module
Event(s) recovered	Front/Rear (1), Side (1)

Comments

imaged direct to module with cable 614 and 387 adaptor

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire crash.
- Airbag ECU data should be used in conjunction with other physical evidence obtained from the vehicle and the surrounding circumstances.
- If the airbags did not deploy or the pretensioners did not operate during an event that meets a specified recording threshold, it is called a Non-Deployment Event. Data from a Non-Deployment Event can be overwritten by a succeeding event that meets the specified recording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event

Events: Front/Rear vs Side vs Rollover - & how many

Side Event Polarity

Which Side were you hit on?

System Status at Event (Most Recent, TRG 2)

Recording Status, Side Crash Info.	Complete
Event Type	Side Crash
TRG Count (times)	2
Recorded Side	Passenger's Side
Previous Crash type	Frontal/Rear
Time from Pre-Crash TRG (msec)	24
Linked Pre-Crash Page	0
Time to Deployment Command, Row 1 (msec)	Not Commanded
Time to Deployment Command, Row 2 (msec)	Not Commanded

- Side contacted is NOT determined by sign.
- Recorded Side is displayed.

New “13 EDR” & “15 EDR”

- 13 EDR Adds lateral Accel data from ACM (“floor sensor”) in ***front/rear*** **events, synchronized to Long DV time 0** (Fixes biggest drawback to 12 EDR),
- Lateral acceleration data table is in meters/sec^2, calculate Delta V from it yourself. For each point, take **(accel/9.81)*(32.2/1.466)*time interval**, and add them to get the cumulative value in **mph**.
- Metric, take accel*3.6*time interval, add them to get the cumulative value in kph.
- See “Toyota 13 EDR” **TEMPLATE** in student materials to do this math for you calculates both Delta V and PDOF.
- Side events are still recorded separately in side wakeups, & record 70ms of side accel, but are NO LONGER NEEDED for analysis if the frontal algorithm triggered and recorded with side data. .
- 15 EDR adds side impact door pressure sensor. Pressure sensor data is NOT used in reconstruction analysis, it is just for quicker side airbag activation.

13 EDR Long DV & Lat Accel

Longitudinal/Lateral Crash Pulse (3rd Prior Event, TRG 1 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	
0	0.0 [0.0]	0.0	
10	-1.1 [-1.7]	4.6	ON
20	-2.0 [-3.3]	7.7	ON
30	-6.6 [-10.6]	41.4	ON
40	-7.8 [-12.5]	96.5	ON
50	-8.6 [-13.8]	19.9	ON
60	-9.1 [-14.7]	-79.7	ON
70	-9.8 [-15.8]	18.4	ON
80	-11.3 [-18.1]	-16.9	ON
90	-12.8 [-20.6]	64.4	ON
100	-13.8 [-22.2]	23.0	ON
110	-15.5 [-25.0]	30.6	ON
120	-17.6 [-28.3]	-49.0	ON
130	-19.8 [-31.8]	13.8	ON
140	-22.7 [-36.5]	24.5	ON
150	-22.3 [-35.9]	-76.6	ON
160	-23.3 [-37.5]	52.1	ON
170	-24.0 [-38.6]	6.1	ON
180	-24.7 [-39.8]	-13.8	ON
190	-25.6 [-41.2]	-12.3	ON
200	-26.1 [-42.1]	9.2	ON
210	-27.0 [-43.5]	29.1	ON
220	-27.5 [-44.2]	26.0	ON
230	-27.7 [-44.6]	19.9	ON
240	-27.9 [-44.9]	7.7	ON
250	-28.1 [-45.2]	4.6	ON

Save as CSV
Integrate to get Δv

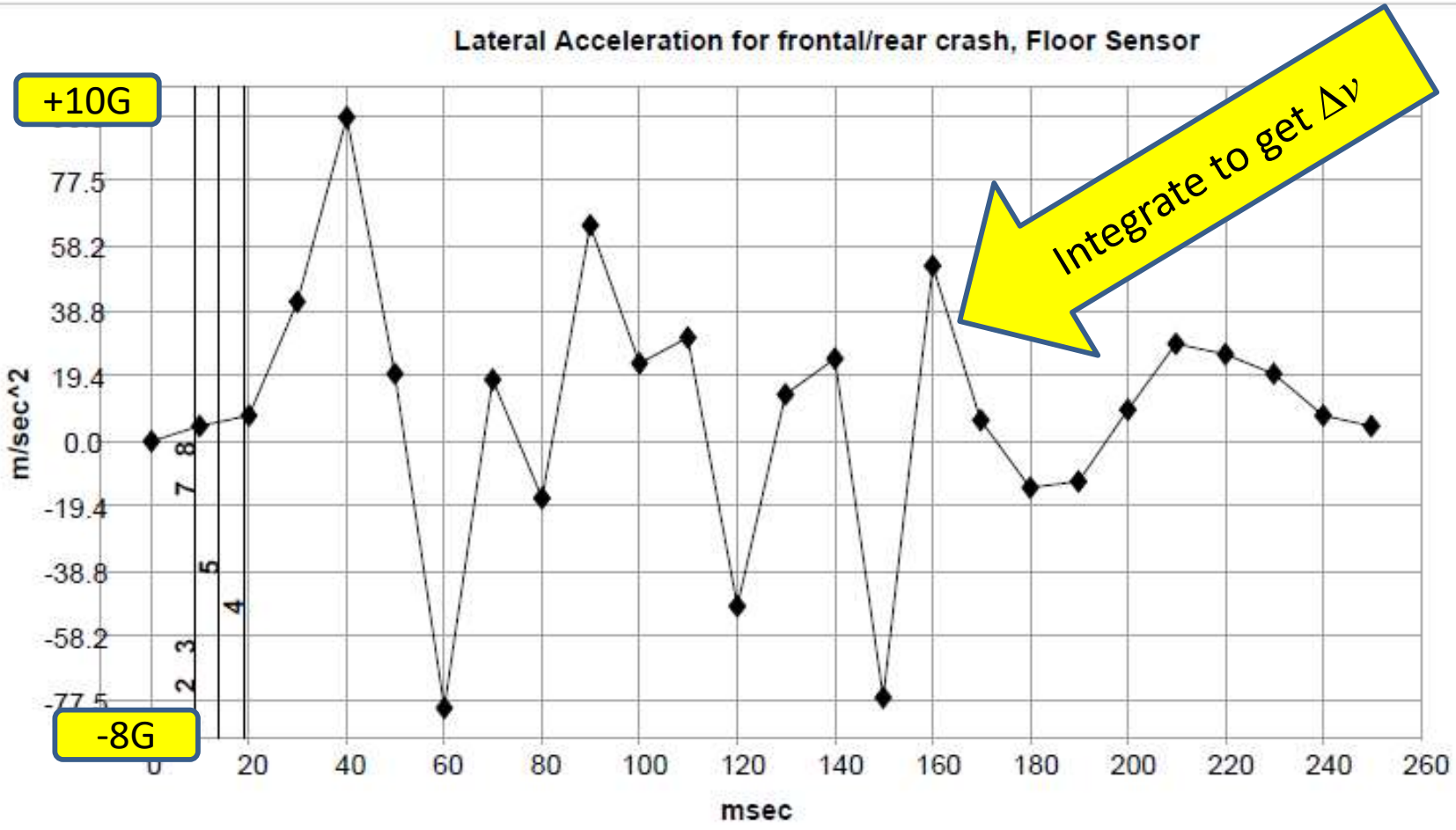
Template in Student Ref Materials

Toyota 13 EDR Lat Accel to Lat DV Template

INSTRUCTIONS
While in CDR save data is CSV format
Copy Columns A thru E from CSV file for 0-250ms
Paste into Column A thru E below. D is metric and is hidden

LONGITUDINAL/LATERAL		PASTE DATA BELOW		CALCULATIONS - IMPERIAL			CALCUALTIONS - METRIC		
Time (msec	Longitudinal Delta-V (MPH)	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	Power Supply Status	Lat DV from Floor Sensor, (MPH)	PDOF in degrees	Total DV	Long DV (KPH)	Lat DV (kph)	Total DV (kph)
0	0	0	ON	0		0.0	0.0	0.0	0.0
10	-1.2	-5.7	ON	-0.1	6.1	1.2	-1.9	-0.2	1.9
20	-3.6	-11.5	ON	-0.4	6.1	3.6	-5.8	-0.6	5.8
30	-6.7	-17.2	ON	-0.8	6.6	6.7	-10.8	-1.2	10.9
40	-9.9	-70.9	ON	-2.4	13.4	10.2	-16.0	-3.8	16.4
50	-13	-113	ON	-4.9	20.6	13.9	-21.0	-7.9	22.3
60	-15.6	-51.7	ON	-6.1	21.2	16.7	-25.1	-9.7	26.9
70	-16.5	-34.5	ON	-6.8	22.5	17.9	-26.5	-11.0	28.7
80	-17.3	-7.7	ON	-7.0	22.0	18.7	-27.9	-11.3	30.0
90	-17.5	-15.3	ON	-7.3	22.8	19.0	-28.1	-11.8	30.5
100	-17.7	-17.2	ON	-7.7	23.6	19.3	-28.4	-12.4	31.1
110	-18	0	ON	-7.7	23.2	19.6	-29.0	-12.4	31.5
120	-18.2	1.9	ON	-7.7	22.9	19.8	-29.2	-12.4	31.8
130	-18.3	0	ON	-7.7	22.8	19.8	-29.5	-12.4	31.9
140	-18.3	0	ON	-7.7	22.8	19.8	-29.5	-12.4	31.9
150	-18.3	3.8	ON	-7.6	22.5	19.8	-29.5	-12.2	31.9
$grey160	$grey0.0	$grey0.0	$greyON	#VALUE!	#VALUE!	#VALUE!	$grey0.0	#VALUE!	#VALUE!

13 EDR lat accel graph from 28 mph (big) longitudinal event



Deployment Time Marker Key

1	Driver Airbag Deployment Time
2	Passenger Airbag Deployment Time
3	Driver/Passenger Pretensioner
4	Driver 2nd Stage Airbag Deployment Time
5	Passenger 2nd Stage Airbag Deployment
6	Driver/Passenger AHR
7	Driver CSA
8	Passenger CSA
9	Rear Window Airbag Deployment Time

17 EDR Event Record Summary
more events, detail on recording status

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	10	Side Crash	0	N/A	Complete	N/A		Complete
1st Prior Event	9	Frontal/Rear/Side	-898	Complete		Complete	Complete	Complete
2nd Prior Event	8	Side Crash	-925	N/A			N/A	Complete
3rd Prior Event	6	Side Crash	N/A		Complete	N/A	N/A	Complete
4th Prior Event	5	Rollover	N/A	Complete	Complete		Incomplete	Incomplete
5th Prior Event	4	Frontal/Rear/Side Crash	N/A	Complete		Complete	Complete	Complete
6th Prior Event	3	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
7th Prior Event	2	Rollover	N/A	Complete	Complete	Complete	Complete	Complete

Pages NOT numbered like Gen3

#7 missing

Lost count due to 7 missing

No precrash for sides

17 EDR –Time in Event Record Summary "Lost" by recovery event #13, 3 key cycles after crash.

Event(s) recovered	Front/Rear/Side Events (2), Side Events (4), Rollover Events (2)

Note Severity of Each Event in the summary

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	13	Side Crash	0	N/A	Complete	N/A	N/A	Complete
1st Prior Event	12	146° Rollover	N/A	Complete	Complete	Complete	Complete	Complete
2nd Prior Event	10	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
3rd Prior Event	9	Frontal/Rear/Side Crash	N/A	Complete	Complete	Complete	Complete	Complete
4th Prior Event	8	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
5th Prior Event	7	200° Rollover	Deployment	Complete	Complete	Complete	Complete	Complete
6th Prior Event	5	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
7th Prior Event	3	Frontal/Rear/Side Crash	-38 DVx, -20 DVy N/A Deployment	Complete	Complete	Complete	Complete	Complete

3 key cycles after 12

-3 DVx, -6DVy

#11 skipped?

Lost at #13

#6 side overwritten?

#4 side overwritten?

#3 Says #2 was not related (so we didn't miss anything important)

17 EDR –Time in Event Record Summary "Lost" by recovery event #13, 3 key cycles after crash. Compare Time Count at Event to find relative time of events

Deployment

Deployment

#3 Big hit -38 DVx

System Status at Event (7th Prior Event, TRG 3)

TRG Count (times)	3	5	7	8	9
Event Type	Frontal/Rear/Side Crash	Side Crash	Rollover	Side Crash	Frontal/Rear/Side Crash
Previous Crash Type	No Event	Side Crash	Side Crash	Rollover	Side Crash
Time from Previous TRG (msec)	32767 or greater	104.5	32.5	916.0	50.0
Time from Time Zero to TRG (msec)	1.5	ON	ON	OFF	46.5
Event Establishment Factor	Frontal Crash	loyment, Driver / Front Airbag Deployment, Passenger	llover CSA Deployment	None	Side Crash, Right
TRG Establishment Factor	Frontal Crash				Side Crash, Right
Freeze Signal	ON				OFF
Freeze Signal Factor					None
Recording Status, Front/Rear and Side Crash Info.	Complete				Complete
Odometer signal (miles [km])	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]
Trip count (times)	4431	4431	4431	4431	4431
Time count (msec)	1,711,500	1,711,600	1,713,500	1,714,300	1,714,400
Time count input system	Normal	Normal	Normal	Normal	Normal

Deployment Command Data (7th Prior Ever

Active Head Restraint, Time to Deploy, Driver (msec)	SNA	Complete	415.0	Complete	
Active Head Restraint, Time to Deploy, Front Passenger (msec)	SNA	0	415.0	0	
Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver (	2.0	0	SNA	0	
Frontal Airbag Deployment, Time to 1st Stage Deployment, Front P	No	None	SNA		
Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec)	12.0		415.0	31901), (B1906),	
Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (m	No		415.0	B1836 1821)	
Pretensioner Deployment, Time to Fire, 1st Seat, Driver (msec)	1.5				
Pretensioner Deployment, Time to Fire, 1st Seat, Passenger (msec	1.5				
Pretensioner Deployment, Time to Fire, 2nd Seat, Driver (msec)	SNA				
Pretensioner Deployment, Time to Fire, 2nd Seat, Passenger (msec	SNA				
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA				
Side Airbag Deployment, Time to Deploy, 1st Seat, Driver (msec)	No				
Side Airbag Deployment, Time to Deploy, 1st Seat, Passenger (mse	140.0				
Side Airbag Deployment, Time to Deploy, 2nd Seat, Driver (msec)	SNA				
Side Airbag Deployment, Time to Deploy, 2nd Seat, Passenger (ms	SNA				
Side Curtain Airbag Deployment, Time to Deploy, Driver (msec)	2.0				
Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec	2.0				

No DTC's yet

3 DTC's

Many DTC's

DTCs Present at Time of Event (7th Prior Ev

Recording Status, Diagnostic	Complete		Complete		
Ignition Cycle Since DTC was Set (times)	0		0		
Airbag Warning Lamp ON Time Since DTC was Set (min)	0		0		
Diagnostic Trouble Codes	None		(B1862), (B1837), (B1832)		

No DTC's yet

DTC's take 2-3 sec to set

10	12	13	
Side Crash	Rollover	Side Crash	
Frontal/Rear/Side Crash	Rollover	No Event	
1,441.0	2,592.0	32767 or greater	
OFF	OFF	OFF	
None	None	None	
45,240 [72,807]	Complete	Invalid	
4431	45,240 [72,807]	4434	
1,715,800	4431	115,900	
Normal	1,718,600	IG	
	Normal		

New **Look** to Precrash Data (same data)

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	29.8 [48]	49.5	24.5	Invalid	2,400	Invalid	OFF
-4.50	32.3 [52]	47.0	24.5	Invalid	2,500	Invalid	OFF
-4.00	34.2 [55]	38.0	20.0	Invalid	2,700	Invalid	OFF
-3.50	35.4 [57]	32.5	16.0	Invalid	2,800	Invalid	OFF
-3.00	36.7 [59]	26.5	13.5	Invalid	2,400	Invalid	OFF
-2.50	37.3 [60]	22.5	11.0	Invalid	2,400	Invalid	OFF
-2.00	37.3 [60]	19.0	8.5	Invalid	2,200	Invalid	OFF
-1.50	38.5 [62]	19.0	6.0	Invalid	1,900	Invalid	OFF
-1.00	37.9 [61]	18.0	5.5	Invalid	1,900	Invalid	OFF
-0.50	37.9 [61]	18.0	5.0	Invalid	1,800	Invalid	OFF
TRG(0)	36.0 [58]	100.0	26.0	Invalid	1,700	Invalid	OFF

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	1.579	-0.98	-4.5	D
-4.50	OFF	OFF	0.00	1.795	0.00	-1.5	D
-4.00	OFF	OFF	0.00	1.292	0.49	0.0	D
-3.50	OFF	OFF	0.00	0.861	0.00	-1.5	D
-3.00	OFF	OFF	0.00	0.718	0.00	-1.5	D
-2.50	OFF	OFF	0.00	-0.072	0.00	-1.5	D
-2.00	OFF	OFF	0.00	0.000	0.00	-1.5	D
-1.50	OFF	OFF	0.00	0.000	-0.49	-1.5	D
-1.00	OFF	OFF	0.00	0.287	0.00	-1.5	D
-0.50	OFF	OFF	0.00	-0.144	0.00	-1.5	D
TRG(0)	OFF	OFF	0.48	-1.507	61.00	-19.5	D

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-5.00	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

17 EDR Event Record Summary
more events, detail on recording status

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	10	Side Crash	0	N/A	Complete	N/A		Complete
1st Prior Event	9	Frontal/Rear/Side	-898	Complete		Complete	Complete	Complete
2nd Prior Event	8	Side Crash	-925	N/A			N/A	Complete
3rd Prior Event	6	Side Crash	N/A		Complete	N/A	N/A	Complete
4th Prior Event	5	Rollover	N/A	Complete	Complete		Incomplete	Incomplete
5th Prior Event	4	Frontal/Rear/Side Crash	N/A	Complete		Complete	Complete	Complete
6th Prior Event	3	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
7th Prior Event	2	Rollover	N/A	Complete	Complete	Complete	Complete	Complete

Pages NOT numbered like Gen3

#7 missing

Lost count due to 7 missing

No precrash for sides

17 EDR –Time in Event Record Summary "Lost" by recovery event #13, 3 key cycles after crash.

Event(s) recovered	Front/Rear/Side Events (2), Side Events (4), Rollover Events (2)

Note Severity of Each Event in the summary

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	13	Side Crash	0	N/A	Complete	N/A	N/A	Complete
1st Prior Event	12	146° Rollover	N/A	Complete	Complete	Complete	Complete	Complete
2nd Prior E	10	Side Cra	N/A	N/A	Complete	N/A	N/A	Complete
3rd Prior Event	9	Frontal/Rear/Side Crash	N/A	Complete	Complete	Complete	Complete	Complete
4th Prior Event	8	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
5th Prior Eve	7	200° Rollover	Deployment	Complete	Complete	Complete	Complete	Complete
6th Pri	5	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
7th Pri	3	Frontal/Rear/Side Crash	-38 DVx, -20 DVy N/A Deployment	Complete	Complete	Complete	Complete	Complete

3 key cycles after 12

-3 DVx, -6DVy

#11 skipped?

Lost at #13

#6 side overwritten?

#4 side overwritten?

#3 Says #2 was not related (so we didn't miss anything important)

17 EDR –Time in Event Record Summary "Lost" by recovery event #13, 3 key cycles after crash. Compare Time Count at Event to find relative time of events

Deployment

Deployment

#3 Big hit -38 DVx

System Status at Event (7th Prior Event, TRG 3)

TRG Count (times)	3	5	7	8	9
Event Type	Frontal/Rear/Side Crash	Side Crash	Rollover	Side Crash	Frontal/Rear/Side Crash
Previous Crash Type	No Event	Side Crash	Side Crash	Rollover	Side Crash
Time from Previous TRG (msec)	32767 or greater	104.5	32.5	916.0	50.0
Time from Time Zero to TRG (msec)	1.5	ON	ON	OFF	46.5
Event Establishment Factor	Frontal Crash	loyment, Driver / Front Airbag Deployment, Passenger	llover CSA Deployment	None	Side Crash, Right
TRG Establishment Factor	Frontal Crash				Side Crash, Right
Freeze Signal	ON				OFF
Freeze Signal Factor					None
Recording Status, Front/Rear and Side Crash Info.	Complete				Complete
Odometer signal (miles [km])	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]
Trip count (times)	4431	4431	4431	4431	4431
Time count (msec)	1,711,500	1,711,600	1,713,500	1,714,300	1,714,400
Time count input system	Normal	Normal	Normal	Normal	Normal

Deployment Command Data (7th Prior Ever

Active Head Restraint, Time to Deploy, Driver (msec)	SNA	Complete	415.0	Complete	
Active Head Restraint, Time to Deploy, Front Passenger (msec)	SNA	0	415.0	0	
Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver (	2.0	0	SNA	0	
Frontal Airbag Deployment, Time to 1st Stage Deployment, Front P	No	None	SNA		
Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec)	12.0		415.0	31901), (B1906),	
Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (m	No		415.0	B1836 1821)	
Pretensioner Deployment, Time to Fire, 1st Seat, Driver (msec)	1.5				
Pretensioner Deployment, Time to Fire, 1st Seat, Passenger (msec	1.5				
Pretensioner Deployment, Time to Fire, 2nd Seat, Driver (msec)	SNA				
Pretensioner Deployment, Time to Fire, 2nd Seat, Passenger (msec	SNA				
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA				
Side Airbag Deployment, Time to Deploy, 1st Seat, Driver (msec)	No				
Side Airbag Deployment, Time to Deploy, 1st Seat, Passenger (mse	140.0				
Side Airbag Deployment, Time to Deploy, 2nd Seat, Driver (msec)	SNA				
Side Airbag Deployment, Time to Deploy, 2nd Seat, Passenger (ms	SNA				
Side Curtain Airbag Deployment, Time to Deploy, Driver (msec)	2.0				
Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec	2.0				

No DTC's yet

3 DTC's

Many DTC's

DTCs Present at Time of Event (7th Prior Ev

Recording Status, Diagnostic	Complete		Complete		
Ignition Cycle Since DTC was Set (times)	0		0		
Airbag Warning Lamp ON Time Since DTC was Set (min)	0		0		
Diagnostic Trouble Codes	None		(B1862), (B1837), (B1832)		

No DTC's yet

DTC's take 2-3 sec to set

10	12	13	
Side Crash	Rollover	Side Crash	
Frontal/Rear/Side Crash	Rollover	No Event	
1,441.0	2,592.0	32767 or greater	
OFF	OFF	OFF	
None	None	None	
45,240 [72,807]	Complete	Invalid	
4431	45,240 [72,807]	4434	
1,715,800	4431	115,900	
Normal	1,718,600	IG	
	Normal		

New **Look** to Precrash Data (same data)

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	29.8 [48]	49.5	24.5	Invalid	2,400	Invalid	OFF
-4.50	32.3 [52]	47.0	24.5	Invalid	2,500	Invalid	OFF
-4.00	34.2 [55]	38.0	20.0	Invalid	2,700	Invalid	OFF
-3.50	35.4 [57]	32.5	16.0	Invalid	2,800	Invalid	OFF
-3.00	36.7 [59]	26.5	13.5	Invalid	2,400	Invalid	OFF
-2.50	37.3 [60]	22.5	11.0	Invalid	2,400	Invalid	OFF
-2.00	37.3 [60]	19.0	8.5	Invalid	2,200	Invalid	OFF
-1.50	38.5 [62]	19.0	6.0	Invalid	1,900	Invalid	OFF
-1.00	37.9 [61]	18.0	5.5	Invalid	1,900	Invalid	OFF
-0.50	37.9 [61]	18.0	5.0	Invalid	1,800	Invalid	OFF
TRG(0)	36.0 [58]	100.0	26.0	Invalid	1,700	Invalid	OFF

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	1.579	-0.98	-4.5	D
-4.50	OFF	OFF	0.00	1.795	0.00	-1.5	D
-4.00	OFF	OFF	0.00	1.292	0.49	0.0	D
-3.50	OFF	OFF	0.00	0.861	0.00	-1.5	D
-3.00	OFF	OFF	0.00	0.718	0.00	-1.5	D
-2.50	OFF	OFF	0.00	-0.072	0.00	-1.5	D
-2.00	OFF	OFF	0.00	0.000	0.00	-1.5	D
-1.50	OFF	OFF	0.00	0.000	-0.49	-1.5	D
-1.00	OFF	OFF	0.00	0.287	0.00	-1.5	D
-0.50	OFF	OFF	0.00	-0.144	0.00	-1.5	D
TRG(0)	OFF	OFF	0.48	-1.507	61.00	-19.5	D

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-5.00	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

17 EDR time data & ped events

System Status at Pedestrian Event (1st Prior Event, TRG 1)

TRG Count (times)	1
TRG Count not for Pedestrian (times)	SNA
Ignition Cycle, Crash (times)	353
Odometer signal (miles [km])	1,195 [1,923]
Trip count (times)	392
Time count (msec)	1,283,900
Time count input system	Normal
Airbag Warning Lamp, On/Off	OFF
Pedestrian Warning Lamp, On/Off	OFF
Time from Previous TRG (msec)	32767 or greater
Time from Pre-Crash to TRG (msec)	300
Time between TRG not for Pedestrian and TRG for Pedestrian (msec)	SNA
TRG Establishment not for Pedestrian, within 5 sec.	ON
Freeze Signal	ON
Recording Status, Pre-Crash	Complete
Recording Status, Crash Info.	Complete
Recording Status, EDR Correlation Info.	Complete

Not present in side events

Similar to Ignition Cycles but slightly higher

Ignition Time on to event

Deployment Command Data (1st Prior Event, TRG 1)

Front Pop-Up Hood, Time to Deploy (msec)	3.0
Rear Pop-Up Hood, Time to Deploy (msec)	3.0
Airbag for Pedestrian Deployment, Time to Deploy (msec)	SNA

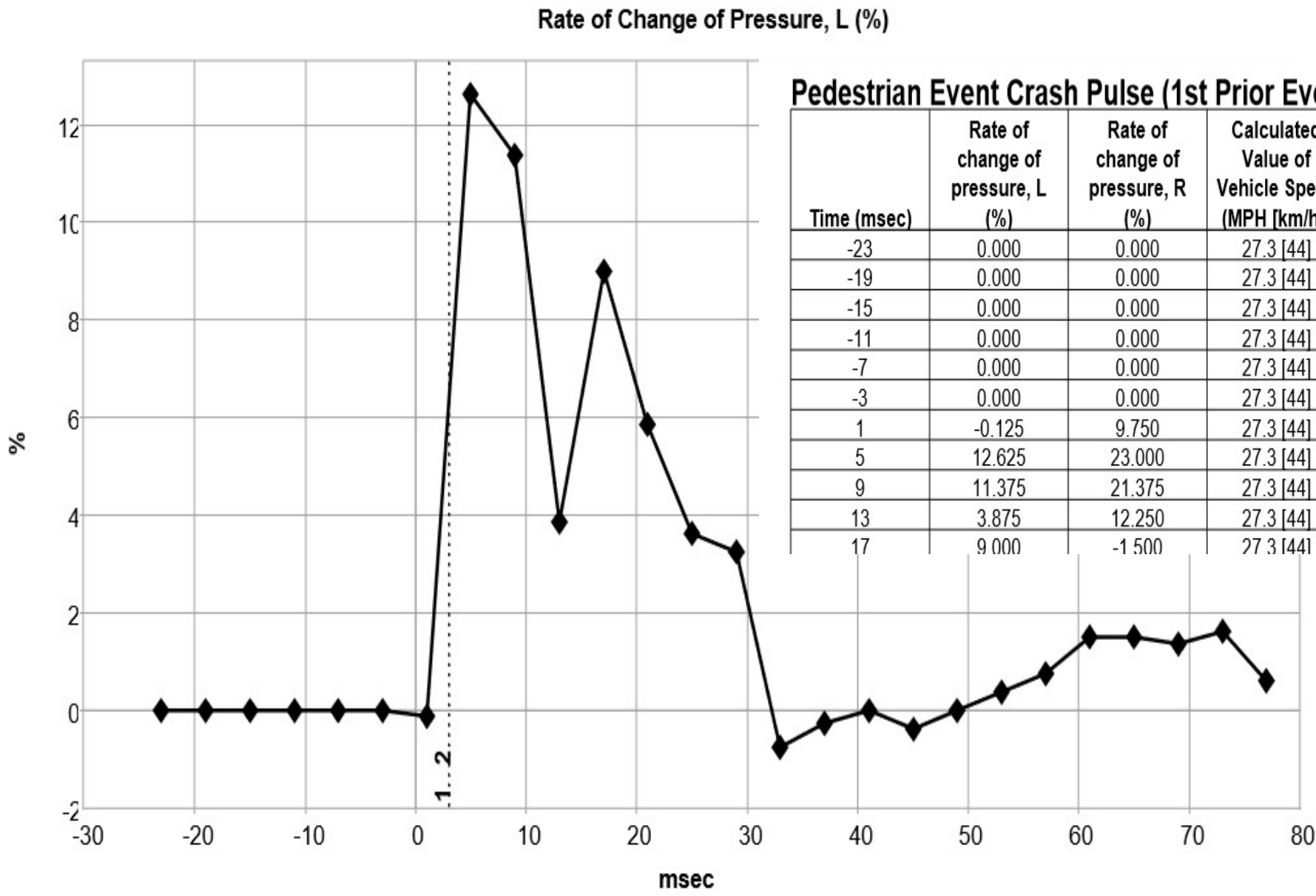
DTCs Present at Time of Event (1st Prior Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Warning ON Time Since DTC was Set, for Pedestrian (min)	0
Diagnostic Trouble Codes	None

Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	1

Data from L/R bumper pressure sensors



Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Time (msec)	Rate of change of pressure, L (%)	Rate of change of pressure, R (%)	Calculated Value of Vehicle Speed (MPH [km/h])
-23	0.000	0.000	27.3 [44]
-19	0.000	0.000	27.3 [44]
-15	0.000	0.000	27.3 [44]
-11	0.000	0.000	27.3 [44]
-7	0.000	0.000	27.3 [44]
-3	0.000	0.000	27.3 [44]
1	-0.125	9.750	27.3 [44]
5	12.625	23.000	27.3 [44]
9	11.375	21.375	27.3 [44]
13	3.875	12.250	27.3 [44]
17	9.000	-1.500	27.3 [44]

2022 Mirai,19+ Lexus LS PPM (Pedestrian)

- PPM memory is integral to ACM but requires downloading a separate file to get the data. Two event spaces.
- Future PPM may be a separate module.
- Help file will identify this, look to see if ACM/PPM use same cable

2021 LS HV (all markets)	ACM	N/A	Data - 19EDR	CDR 900	CDR 900 - click here	Cable# 827 CDR 900 - click here (use listed cable above and no ACM adapter)	Under Center Console
2019 - 2021 (non-HV)	PPM (pop-up hood) *if equipped*	N/A	Data - 17EDR	CANplus or CDR 900	CANplus - F00K108287 *- for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to vehicles sold outside the*	CANplus - F00K108387 & Cable#827 CDR 900 - click here (use listed cable above and no ACM adapter)	*Note: if equipped with PPM, its EDR data is stored in the ACM but requires separate downloads for each system*

Pedestrian Protection Events VIN_PPM

EDR Device Type	Pedestrian Protection Module
Event(s) recovered	Pedestrian Events (2)

Read separate from ACM

System Status at Pedestrian Event (1st Prior Event, TRG 1)

TRG Count (times)	1
TRG Count not for Pedestrian (times)	SNA
Ignition Cycle, Crash (times)	353
Odometer signal (miles [km])	1,195 [1,923]
Trip count (times)	392
Time count (msec)	1,283,900
Time count input system	Normal
Airbag Warning Lamp, On/Off	OFF
Pedestrian Warning Lamp, On/Off	OFF
Time from Previous TRG (msec)	32767 or greater
Time from Pre-Crash to TRG (msec)	300
Time between TRG not for Pedestrian and TRG for Pedestrian (msec)	SNA
TRG Establishment not for Pedestrian, within 5 sec.	ON
Freeze Signal	ON
Recording Status, Pre-Crash	Complete
Recording Status, Crash Info.	Complete
Recording Status, EDR Correlation Info.	Complete

Synch to ACM

Deployment Command Data (1st Prior Event, TRG 1)

Front Pop-Up Hood, Time to Deploy (msec)	3.0
Rear Pop-Up Hood, Time to Deploy (msec)	3.0
Airbag for Pedestrian Deployment, Time to Deploy (msec)	SNA

DTCs Present at Time of Event (1st Prior Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Warning ON Time Since DTC was Set, for Pedestrian (min)	0
Diagnostic Trouble Codes	None

Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	1

VCH Freeze Frame Data

- New feature in 2013+ Rav 4 and 2014+ Highlander & Lexus IS, 2015+ Lexus NX & RC, "**V**ehicle **C**ontrol **H**istory"
- Triggers EDR-style recording if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion. 26 triggers including crash avoidance, ESC.
- Accessible only with Toyota Techstream service tool – not part of Bosch CDR system. Annual s/w subscription or 2 day subs available.
- CDR900 or Nexiq works as pass thru device same as Toyota Mongoose hardware
- Tabular Data exports as ".csv" –
- Space in memory for MANY events – can show a history of erratic driver behavior
- Details in SAE 2016-01-1495 Appendix D. Also **2019-01-0632** "**Reconstructing Vehicle Dynamics from On-Board Event Data**" **discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events).**

VCH Freeze Frame Data

- Early VCH systems have 26 different triggers and record only tabular EDR-style precrash data
- TSS-P & TSS-C, aka “1.0” systems, can record pictures if and only if the automatic emergency braking system engages (applies the brakes for you). A warning is not enough to record pictures. CHECK: perform CDR download FIRST. Look for deceleration with no driver input: i.e. no brake switch = on, no brake oil pressure, but speed drops more than idling down
- TSS “2.0” and “2.5” systems also record pictures in the event of a crash. See application chart that follows (best info available, not guaranteed).

TOYOTA/LEXUS Vehicles with Vehicle Control History Data and LSS/TSS with Techstream Data

		2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
LEXUS	ES				LSS	LSS	LSS	LSS 2.0	LSS 2.0	LSS 2.0	LSS+ 2.5	LSS+ 2.5	LSS+ 2.5
	GS				LSS	LSS	LSS	LSS	LSS	discontinued			
	GX								LSS	LSS	LSS	LSS 2.0	LSS 3.0
	IS		250 and 350			LSS	LSS	LSS	LSS	LSS 2.0	LSS+ 2.5	LSS+ 2.5	LSS+ 2.5
	LC						LSS	LSS	LSS	LSS	LSS	LSS	LSS+ 2.5
	LS						LSS	LSS 2.0	LSS 2.0	LSS 2.0	LSS+ 2.5	LSS+ 2.5	LSS 3.0
	LX						LSS	LSS 2.0	LSS 2.0	LSS 2.0	LSS+ 2.5	LSS+ 2.5	LSS+ 2.5
	NX						LSS		LSS 2.0	LSS 2.0	LSS 3.0	LSS 3.0	LSS 3.0
	RC						LSS	LSS	LSS	LSS	LSS	LSS+ 2.5	LSS+ 2.5
	RX				LSS	LSS	LSS		LSS 2.0	LSS 2.0	LSS 2.0	LSS 3.0	LSS 3.0
	UX							LSS 2.0	LSS 2.0	LSS 2.0	LSS 2.0	LSS+ 2.5	LSS+ 2.5
TOYOTA	4Runner								TSS-P	TSS-P	TSS-P	TSS-P	TSS-P
	Avalon				TSS-P	TSS-P	TSS-P	TSS-P	TSS-P	TSS-P	TSS+ 2.5		
	C-HR						TSS-P	TSS-P	TSS-P	TSS+ 2.5	TSS+ 2.5		
	Camry						TSS-P	TSS-P	TSS-P	TSS+ 2.5	TSS+ 2.5	TSS+ 2.5	TSS+ 2.5
	Corolla						TSS-P	TSS 2.0	TSS 2.0	TSS 2.0	TSS 2.0	TSS 3.0	TSS 3.0
	Highlander							TSS-P	TSS 2.0	TSS+ 2.5	TSS+ 2.5	TSS+ 2.5	TSS+ 2.5
	Land Cruiser				TSS-P	TSS-P	TSS-P	TSS-P	TSS-P	TSS-P			
	Mirai							TSS 2.0	TSS 2.0	TSS+ 2.5	TSS+ 2.5	TSS+ 2.5	TSS 3.0
	Prius			C and P	and Prius	and Prime			TSS-P	TSS 2.0	TSS 2.0	TSS 3.0	TSS 3.0
	RAV4						TSS-P	TSS 2.0	TSS 2.0	TSS 2.0	TSS 2.0	TSS+ 2.5	TSS+ 2.5
	Sequoia						TSS-P	TSS-P	TSS-P	TSS-P	TSS-P		
	Sienna								TSS-P				
	Tacoma						TSS-P	TSS-P	TSS-P	TSS-P	TSS-P	TSS-P	TSS 3.0
	Tundra						TSS-P	TSS-P	TSS-P	TSS-P	TSS+ 2.5	TSS+ 2.5	TSS+ 2.5
	Venza												
	Yaris												
SCION	iM												

Accessing VCH data

- Buy short term Subscription from Toyota
- Software changing -Techstream to Global GTS+
- Techstream discontinued Sept 2024
- CDR 900 still works as VIM but you must select it and input the serial number of your 900
- Users are reporting problems finding everything techstream had in GTS+ - one report required a screen shot for each time stamp instead of one file with it all in it
- Be sure to save PDF's of "VDAS" file before subscription expires – VDAS will not open without subscription

Link to Updated GTS how-to by Brad Muir



GTS+ User Guide for Distribution - 10-2024 Brad Muir.pdf

Detailed VCH Data Per Event



Tips Using VCH Data for Ped Hits

- Use date/time feature and key cycles to find your event. Beware if user sets clock wrong the time will be wrong.
- Typically look for **last** recorded events but often there are postcrash recovery events after the event of interest
- Look for sudden accel release, **sudden braking**, sudden steering avoidance to identify reaction to ped – brake to stop?
- ***Be sure to download CSV file*** and NOT just PDF's- PDF's do not contain all timing info and all data that CSV does
- Look at event type – sudden braking, PCS operation
- Look for ignition on timer in ms to help synchronize events

Images from the Toyota FCM

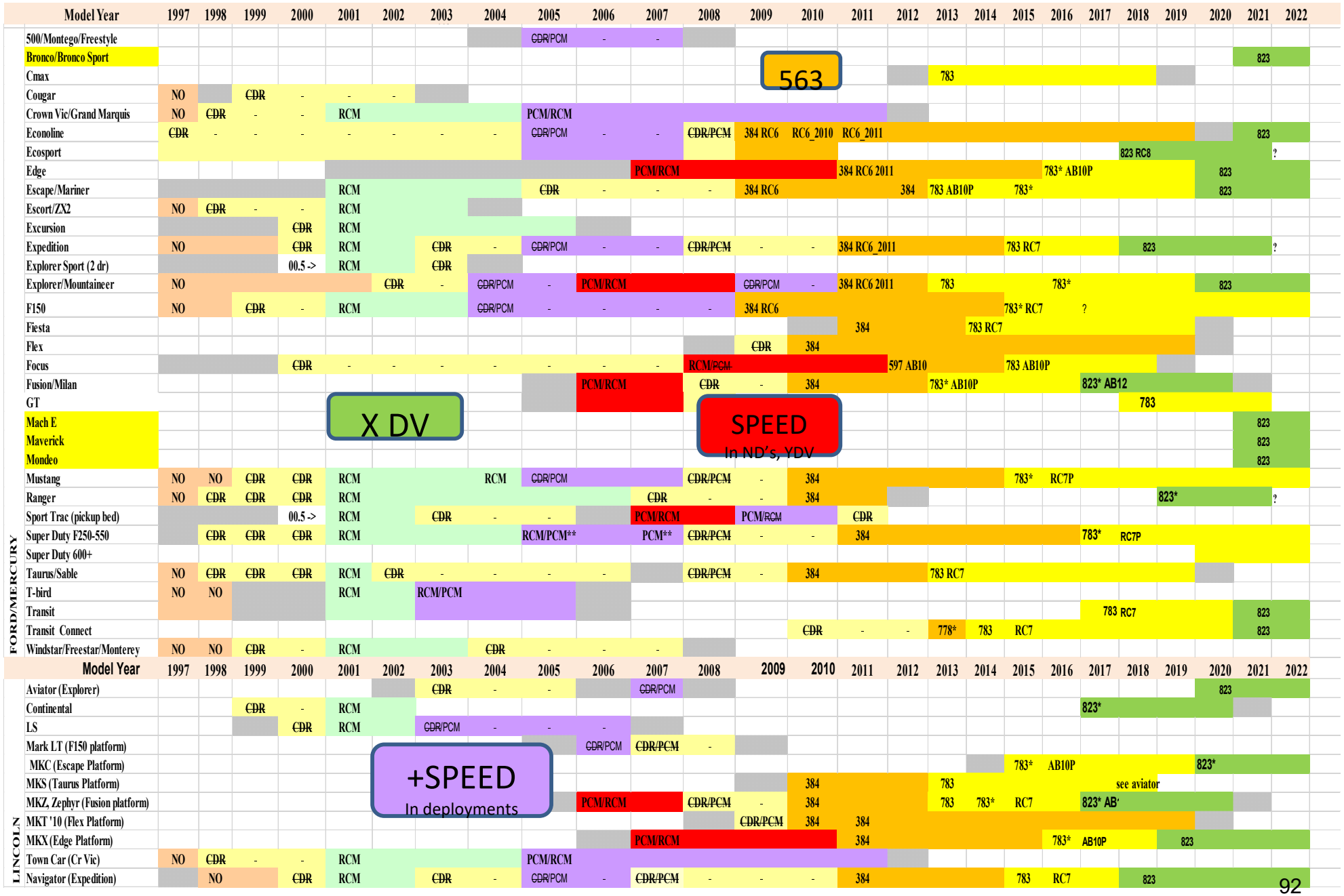


Event Data from Sudden Braking Event

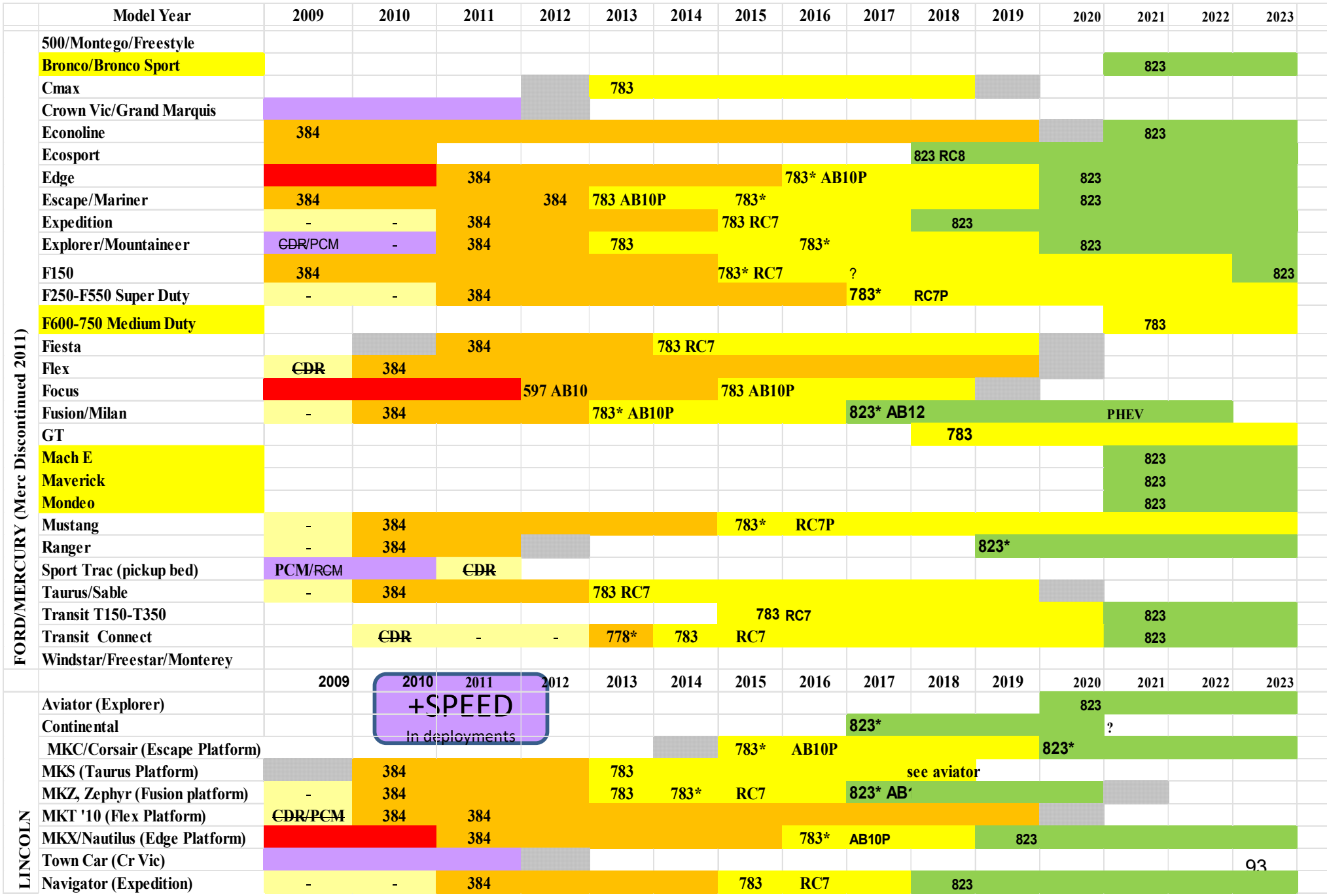
Date/Time Stamp	Time relative to stop	Vehicle Speed [km/h]	Speed MPH	Accelerator Opening Ratio [%]	Engine RPM Data [rpm]	Cruise Control	Shift Position Signal	Brake SW	Brake On=1 Off=0	Brake Oil Pressure [MPa]	Yaw Rate Sensor Signal [deg/s]	Longitudinal G Sensor Signal	Long g's	Lateral G Sensor Signal [m/s^2]	Lat g's	Throttle Opening Ratio [%]	Steering Signal [deg]
2021-11-11-20:05:49.650	-4.65	65	40.3	16.5	1200	0	1	OFF	0	0	-2.928	-0.14356	-0.01462	-0.57424	-0.0586	8.5	-4.5
2021-11-11-20:05:49.800	-4.5	65	40.3	17.5	1200	0	1	OFF	0	0	-2.928	-0.14356	-0.01462	-0.3589	-0.03662	6	-4.5
2021-11-11-20:05:49.950	-4.35	65	40.3	0	1200	0	1	OFF	0	0	-3.416	-0.14356	-0.01462	-0.50246	-0.05127	4.5	-4.5
2021-11-11-20:05:50.100	-4.2	64	39.68	0	1200	0	1	ON	100	6.24	-3.416	-6.89088	-0.70172	-0.50246	-0.05127	0	-9
2021-11-11-20:05:50.200	-4.05	59	36.58	0	1100	0	1	ON	100	10.752	-4.392	-8.9725	-0.9137	-0.50246	-0.05127	0	-18
2021-11-11-20:05:50.250	-3.9	56	34.72	0	1100	0	1	ON	100	11.04	-3.904	-8.75716	-0.89177	-0.93314	-0.09522	0	-22.5
2021-11-11-20:05:50.400	-3.75	55	34.1	0	1200	0	1	ON	100	10.752	-6.832	-8.9725	-0.9137	-0.93314	-0.09522	0	-27
2021-11-11-20:05:50.550	-3.6	46	28.52	0	1000	0	1	ON	100	10.608	-7.808	-8.75716	-0.89177	-1.4356	-0.14649	0	-40.5
2021-11-11-20:05:50.700	-3.45	43	26.66	0	900	0	1	ON	100	10.896	-9.272	-8.9725	-0.9137	-2.29696	-0.23438	0	-58.5
2021-11-11-20:05:50.850	-3.3	39	24.18	0	800	0	1	ON	100	10.8	-11.224	-8.9725	-0.9137	-3.01476	-0.30763	0	-63
2021-11-11-20:05:51.000	-3.15	36	22.32	0	800	0	1	ON	100	10.368	-12.688	-8.75716	-0.89177	-2.5123	-0.25636	0	-67.5
2021-11-11-20:05:51.150	-3	32	19.84	0	900	0	1	ON	100	8.88	-14.152	-8.9725	-0.9137	-1.57916	-0.16114	0	-67.5
2021-11-11-20:05:51.300	-2.85	28	17.36	0	800	0	1	ON	100	4.656	-13.664	-8.32648	-0.84791	-1.4356	-0.14649	0	-67.5
2021-11-11-20:05:51.450	-2.7	25	15.5	0	800	0	1	ON	100	4.128	-11.712	-5.95774	-0.60669	-0.64602	-0.06592	0	-63
2021-11-11-20:05:51.600	-2.55	22	13.64	0	700	0	1	ON	100	3.696	-8.784	-4.59392	-0.46781	-0.14356	-0.01465	0	-58.5
2021-11-11-20:05:51.750	-2.4	19	11.78	0	700	0	1	ON	100	4.224	-7.32	-5.09638	-0.51898	0.14356	0.014649	0	-54
2021-11-11-20:05:51.900	-2.25	17	10.54	0	700	0	1	ON	100	3.168	-6.344	-4.09146	-0.41665	0.3589	0.036622	0	-49.5
2021-11-11-20:05:52.050	-2.1	15	9.3	0	700	0	1	ON	100	2.736	-4.88	-3.87612	-0.39472	0.64602	0.06592	0	-36
2021-11-11-20:05:52.200	-1.95	13	8.06	0	700	0	1	ON	100	2.784	-2.928	-3.66078	-0.37279	0.86136	0.087894	0	-18
2021-11-11-20:05:52.350	-1.8	11	6.82	0	700	0	1	ON	100	2.784	-1.464	-3.66078	-0.37279	0.78958	0.080569	0	-4.5
2021-11-11-20:05:52.500	-1.65	9	5.58	0	600	0	1	ON	100	2.304	-0.488	-3.08654	-0.31431	0.50246	0.051271	0	4.5
2021-11-11-20:05:52.650	-1.5	8	4.96	0	600	0	1	ON	100	1.776	0	-2.58408	-0.26314	0.64602	0.06592	0	18
2021-11-11-20:05:52.800	-1.35	7	4.34	0	600	0	1	ON	100	1.392	0.488	-1.93806	-0.19736	0.78958	0.080569	0	31.5
2021-11-11-20:05:52.950	-1.2	6	3.72	0	600	0	1	ON	100	1.248	0.976	-1.7945	-0.18274	0.64602	0.06592	0	40.5
2021-11-11-20:05:53.100	-1.05	5	3.1	0	600	0	1	ON	100	1.056	0.976	-1.65094	-0.16812	0.43068	0.043947	0	54
2021-11-11-20:05:53.250	-0.9	4	2.48	0	600	0	1	ON	100	0.864	1.464	-1.29204	-0.13157	0.50246	0.051271	0	63
2021-11-11-20:05:53.400	-0.75	3	1.86	0	600	0	1	ON	100	0.864	0.976	-1.29204	-0.13157	0.57424	0.058596	0	76.5
2021-11-11-20:05:53.550	-0.6	3	1.86	0	600	0	1	ON	100	1.104	1.464	-1.36382	-0.13888	0.3589	0.036622	0	85.5
2021-11-11-20:05:53.700	-0.45	2	1.24	0	600	0	1	ON	100	1.104	0.488	-1.50738	-0.1535	0.21534	0.021973	0	90
2021-11-11-20:05:53.850	-0.3	2	1.24	0	600	0	1	ON	100	1.008	0.488	-1.22026	-0.12426	0.43068	0.043947	0	90
2021-11-11-20:05:54.000	-0.15	1	0.62	0	600	0	1	ON	100	0.864	0.488	-1.0767	-0.10964	0.28712	0.029298	0	94.5
2021-11-11-20:05:54.150	0	0	0	0	600	0	1	ON	100	0.528	0	-0.57424		0.3589	0.036622	0	94.5
2021-11-11-20:05:54.300	0	0		0	600	0	1	ON	100	0.576	0	-0.64602		0.3589	0.036622	0	99

FORD

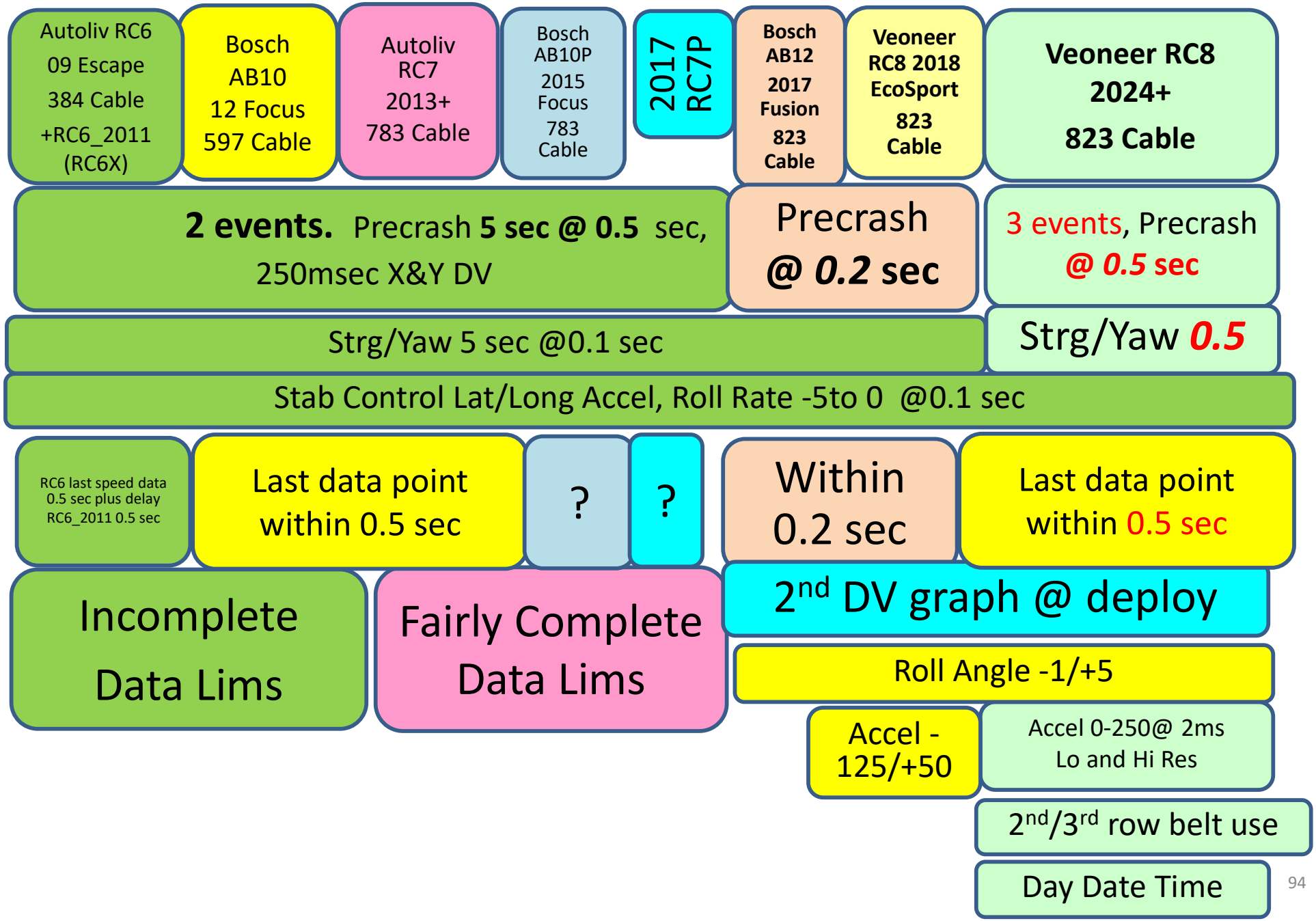
Ford CDR 21.3 Coverage



Ford 09+ CDR 23.2 Coverage 69%



Ford Part 563 EDR Subfamilies



Ford Occupant Size Classification

Pre-Crash Data -1 sec (First Record)

Ignition cycle, crash	359
Frontal air bag warning lamp, on/off	OFF
Occupant size classification, front passenger (Child size Yes/No [Hex value])	No [$08]
Safety belt status, driver	Driver Buckled

- The Hex Value provides additional information than just child yes/no
- The Decoder is different for each supplier and major module type
- Look at end of data limitations for module type – Autoliv=RC5, RC6, RC7; Bosch=AB9, AB10

RC6/7	AB10	AB12	RC8	Classification*
$01	$00	$00	$00	Empty
$02	$01	$05	$02	Child or Occupied Disable
$04	$05	$04	$01	Indeterminate (gray zone between child and adult)
$08	$02	$01	$03	Occupied Enable Small Adult (typical 110 lb 5th female)
$10	$03	$02	$04	Occupied Enable Medium (typical 172 lb 50th male)
$20	$04	$03	$05	Occupied Enable Large Adult (typical 223 lb 95th male)

* Based on communications with Ford sent to EDRFORD@FORD.COM

Stability Control System Data

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record)

Times (sec)	Steering Wheel Angle (degrees)	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)	Stability Control Yaw Rate (deg/sec)	Stability Control Roll Rate (deg/sec)
	Not all have it				Not all have it
- 5.0	0.3	0.012	0.136	-0.02	0.92
- 4.9	0.5	0.024	0.14	0.29	-0.4
- 4.8	0.6	0.0	0.131	0.04	0.52
- 4.7	Polarity Left/CCW=Positive		0.104	CCW=POSITIVE	0.32
- 4.6	0.3	0.007	0.094	0.18	1.8
- 4.5	0.3	-0.002	0.089	0.22	0.32
- 4.4	0.5	0.028	USE TO GET DRAG FACTOR AND SPEED CHANGE	0.25	-1.36
- 4.3	0.6	-0.012		USE TO GET HEADING CHANGE AND SLIP ANGLE	0.84
- 4.2	-0.1	USE TO GET SIDE MOVEMENT			USE TO GET ROLL ANGLE
- 4.1	-0.4				
- 4.0	-0.2				

Stability Control System Data

- SAE 2014-01-0504 **Accuracy and Timing of 2013 Ford Flex Event Data Recorders includes a validation of stability control system data**
- **An analysis template is provided in your student reference materials.**
  - **Get speed at 0.1 sec intervals from a starting stable speed and longitudinal accel**
  - **Get cumulative yaw angle from yaw rate**
  - **Get sideways movement from lat accel**
  - **Get change in approach angle**

Ford Global Real Timer

Added for AB12/ RC7P/RC8

Timer starts when battery is connected and Body Control Module is plugged in. Timer stops if power is disconnected from BCM and starts again when reconnected. Timer is sent to ACM over CAN bus.

Month of build is on door jamb sticker (exact day/time is not)

Can be used to tie your recording to your crash

Great for knowing a more precise time between events when not within 5 seconds

Take VIN to Ford Dealer, ask for specific build day if needed.

Pre-Crash Data -1 sec (First Record)

Ignition cycle, Crash	1,782
Frontal Air Bag Warning Lamp, On/Off	Off
Safety Belt Status, Driver	Belted
Seat Track Position Switch, Foremost, Status, Driver	Rearward
Seat Track Position Switch, Foremost, Status, Front Passenger	Rearward
Safety Belt Status, Front Passenger	Unbelted
Brake Telltale	Off
ABS Telltale	Off
ESC/TC Telltale	Off
ESC/TC Off Telltale	Default Mode
Powertrain Wrench Telltale	Off
Powertrain Malfunction Indicator Lamp (MIL) Telltale	Unchanged Off
Global Real Time (seconds)	47,951,087

Pictures stored in Fords?

- Ford has not made any pictures available from collision avoidance systems. It is not known what is stored if anything.
- The Ford “Blue Cruise” semi-autonomous driving system on limited access highways allegedly stores photos that can be requested through Ford legal.

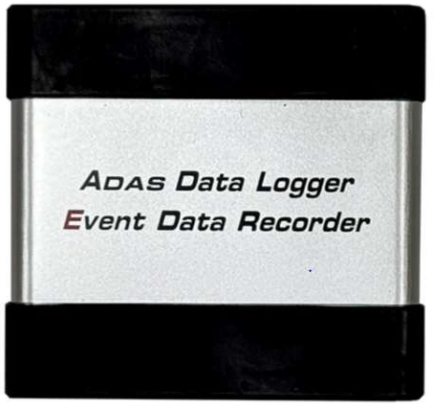
KIA / HYUNDAI

KIA/HYUNDAI

- KIA & HYUNDAI released the ability to read ACM's in vehicles sold in USA and built after 9/1/2012 models to their dealer scan tool maker, Global Information Technologies (GIT).
- 2013+ MY data meets & exceeds requirements of Part 563.
- 2013-2024 had separate kit for each brand and brand specific software, Kia and Hyundai/Genesis.
- Feb 2025 **new "ADL/EDR" 2.0 interface** is capable of doing either brand if you buy the both software subscriptions.
- New Version 2.0 software came out Feb 2025, replaced V 1.0 level 65 Hyundai/level 61 Kia. Software will add capability to do "ADL" Advanced Data Logger soon.

Hyundai/Kia EDR Tools

- New ADL/EDR interface U.S. DLC only kits from Crash Data Group **$5550**
  (software annual subscription $900/brand not included)



- 1ea of 24 DTM cables #1-#30 in a case $7650 from CDG
  (no CFCI adapter)

- OLD interfaces still work with new EDR software but for cables 29-30 require separate CAN-FD (Flexible Data Rate) DTM adapter $1,400

- GIT Contact in transition – last known as (888)549-4977
- Scott Baker **crash@crashdatagroup.com** (800) 280-7940 moving warehouse March 2025 from Temecula CA to Washington state

Additional Info

1. Old K/H Interface modules are sold as brand specific but are identical in function, but you had to buy both kits to get software for both. Software updates won't download unless the **correct brand interface module** is attached to your laptop and the ID & password keyed to that interface are used. US Kit update request must come from within US borders (VPN not honored).

Officially Supported 2013+ Models Software 2.0 or 1.0 V65H/61K

Model	Model Years														
KIA	2010	2011	2012	2013*	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
CADENZA (Azera sister)					19(VG)	19	19	13(YG)	13	13	13	13			
EV6													29+CFCI (CV)		
FORTE	10	10	10	10 (TD)	6 (YD)	6	6	12(YD,YDM)	12	21 or 28 (BDm)				21	
K5												26(D3LA)			
K900						11 (KH)	11	11	11	21 or 28 (RJ)					
NIRO (DE EV, DE HEV, DE PHEV)								21(DE)	21	21	21	21	21	30+CFCI (SG)	
OPTIMA	No	4	4	4 (TF/QF)	4	4	14 (JF,JFA)*	14	14	14	14				
OPTIMA HYBRID	No	4	4	4 (TF HEV)	4	4	4	23(JF HEV)	23	23	23				
RIO	No	No	8	8(UB)	8	8	8	8	21 (SC)	21	21	21	21	21	
RONDO (Canada only 2011 MY+)	No	No	9	(not officially supported since non-US, read as a Kia Rio)											
SEDONA	No	No	No	No	10(VQ)	12(YP)	12	12	12	12	12	12	29+CFCI (KA4)		
SELTOS												28(SP2)	28	28	
SORENTO	No	3	3	3(XM)	3,7(XM)	3,7	13 (UMA)	13	13	13	13	29+CFCI (MQ4)		29	
SOUL	1 see note 1	1 see note 1	1	1(AM)	12(PS)	12(PSEV)	12	12	12	12,26(SK3 EV)	26 (SK3)	26	26	26	
SPORTAGE (SL)	No	Yes	2	2(SL)	2	2	2	13(QL)	13	13	13	13	30+CFCI (SG)		
STINGER (CK)									21 (CK)	21	21	21	21	21	
TELLURIDE (ON)											25(ON)	25	25	25	
HYUNDAI	2010	2011	2012	2013*	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
ACCENT (RB)	No 10	No?	5 see note 2	5(RB)	5	5	5	5	21(HC)	21	21	21	21	21	
AZERA (HG)		likely	Yes	7(HG)	7	7	7	7							
ELANTRA Coupe(JK), Sedan(UD/MD	No	Yes	8	8(JK/UD/MD)	8	8	8	21(AD, ADA)	21	21	21	28 (CN7)			
ELANTRA GT (GD) (aka "i30")	old model		new model	6/12(GD)	6,12	6,12	6,12	6,12	21(PD)	21	21	21			
EQUUS (VI)(stretch genesis)	?	yes?	yes?	9,11(VI)	9,11(VI)	9,11(VI)	9,11(VI)								
GENESIS (Coupe BK, Sedan BH/DH)	9 shave keyway	9 shave keyway	?	9(BK, BH)	9	13(DH)	13		2018	2019	2020	2021	2022	2023	
GENESIS GV60													29+CFCI (JW1)		
GENESIS G70	YELLOW = COULD BE EITHER OF 2 CABLES, SEE WHICH FITS									21,25(IK)	21,25	21,25	21,25	21,25	
GENESIS GV70													29+CFCI (JK1)		
GENESIS G80								13(DH)	13	13	13	29+CFCI (RG3)			
GENESIS GV80											29+CFCI (JX1)				
GENESIS G90								13(HI)	13	13	13	13	13	30+CFCI (RS4)	
IONIQ EV and HEV								21(AE)	21	21	21	21	21		
IONIQ 5													29+CFCI (NE EV)		
KONA (i20, Rio sister)small SUV									24 (OS)	24	24	24	24	24	
KONA EV										27 (OS EV)	27	27	27	27	
NEXO										26 (FE)	26	26	26	26	
PALISADE											25(LX2)	25	25	25	
SANTA CRUZ				2013*	2014	2015	2016	2017	2018	2019	2020		30+CFCI (NX4A OB)		
SANTE FE	9 shave keyway note 3	9 shave keyway	9 shave keyway	7(NC,DMA)	7	7	7	7	7	25 (TMA)	25	25	25	25	
SONATA		4	?	(YFA, YF HEV	4	14 (LF) 4 (HEV)	14 (LF HEV)	14	14	14	25(DN8A)	25	25	25	
TUCSON	Yes	Yes	2	2(LM)	2	2	13(TL)	13	13	13	13	13	30+CFCI		
VELOSTER coupe	out of productio		?	5(FS)	5	5	5	5	5	14 (JS)	14				
VELOSTER N										28(JSN)	25	25	25	25	
VENUE											30(QX)	30			
Gold = 2013 Part 563 Compliant		K/H Common Cables		2013*	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024

Not published yet – use 2.0 software

Pre-2013 Kia Hyundai may

Model	Vehicle Size, Descrip	Model Years 2010	2011	2012	2013*
KIA		Cable #	Cable #	Cable #	Cable #
CADENZA (VG, 2014) (Azera sister)	112.0" WB	YG requires cable 13 per software			
FORTE (TD 2013, YD 2014))	104.3"WB, compact	10	10	10	10 (TD)
K900 (KH 2015)	119.9"WB				
NIRO					
OPTIMA (2013 TF QF, 2016 JF, JFA)	110.0" WB sedan	No	4	4	4 (TF/QF)
OPTIMA HYBRID	110.0" WB sedan	No	4	4	(TF HEV)
RIO (UB)	94.5"WB, compact	No	No	8	8(UB)
RONDO (Canada only 2011 MY+)		No	out of production		
SEDONA (VQ '14, YP '15)	113.8/118.9" WB miniv	No	No	No	No
SORENTO (XM to 2015)(UMA 2016)	106.3" WB, crossover	No	3	3	3(XM)
SOUL (AM 2013, PS 2014+)	100.4" WB	1 see note 1	1 see note 1	1	1(AM)
SPORTAGE (SL)	103.9" WB, crossover	No	Yes	2	2(SL)
HYUNDAI					
ACCENT (RB)	94.5"WB, compact	No 10	No?	5 see note 2	5(RB)
AZERA (HG)	109.4"WB, sedan		likely	Yes	7(HG)
ELANTRA Coupe(JK), Sedan(UD/MD)	104.3"WB, compact	No	Yes	8	8(JK/UD/MD)
ELANTRA GT (GD) (aka "i30")	former Elantra Touring	old model		new model	6(GD)
EQUUS (VI)(stretch genesis)	119.9" WB, luxury seda	?	yes?	yes?	9(VI)
GENESIS (Coupe BK, Sedan BH/DH))	115.6" WB, luxury seda	9 chave keyway	9 chave keyway	?	9(BK, BH)
GENESIS G90					
SANTE FE (Sport DMA, NC)	106.3" WB, crossover	9 chave keyway note 3	9 chave keyway	9 chave keyway	7(NC,DMA)
SONATA (YF sedan, YF HEV)	110.0" WB sedan		4	?	4(YF)
TUCSON (LM to 2015)(TL in 2016)	103.9" WB, crossover	Yes	Yes	2	2(LM)
VELOSTER coupe (FS)	104.3"WB, compact	out of productior		?	5(FS)

Kia/Hyundai Summary

- Different access tool, different format – but same minimum data as other 563 modules
- Lateral Delta V, Steering, RPM from Table 2
- File only saveable as PDF - Cannot reinterpret in later software – must preserve module and re-read raw data if changes to software occur.
- PDF file is officially secure but security is easily broken by software anyone can purchase.
- Tool may read 2010-2012 models, speed and Delta V reliable per SAE 2014-01-0502& SAE 2015-01-1445, but admissibility must be supported with data.
- Throttle, brake on off and driver belt status have subsequently been determined to be correct.

HONDA

Honda 24.2 CDR Coverage

Mostly US, no Euro until 2023

Model	Location	2011	2012*	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
ACURA		Last 3 digits of Bosch CDR cable number					Gen 2		Gen 3			Gen 4				
RL			547													
TL			547													
TSX			547													
ZDX			547													
MDX			547		387/789			387/810				?	846			
NSX								387/789								
RDX				547			387/789		387/825							
ILX				547			387/789									
RLX					387/789											
TLX						387/789				387/810		846				
INTEGRA														846		
HONDA			2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
ACCORD				387/789			387/810		387/825						846	
CIVIC 2 DR			547				387/810						846			
CIVIC 4 DR			547	387/789			387/810						846			
CLARITY												387/810				
CROSSTOUR			547													
CR-V			547				387/789	387/810					846		846	
CR-Z			547													
FCX Clarity								387/810								
FIT		EV	547			387/789		387/810					846			
HR-V							387/789		387/810				846			
INSIGHT			547							387/810						
ODYSSEY			547		387/789			387/810							846	846
PASSPORT										387/810						
PILOT			547				387/789	387/810						846		
RIDGELINE			547					387/810								

2011-
NO time series EDR
But may have Delta V single value

Honda only assists law enforcement voluntarily

The source data for thi… …rsion 24.2
Always consult the latest help file for the most accurate coverage
* Some 2012 models ot covered

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GEN 4 New Data Elements

System Status at Event (Impact Event Record 1)

Seat track position switch, foremost, status, driver	No
Safety belt status, left second row passenger	Off
Safety belt status, middle second row passenger	Off
Safety belt status, right second row passenger	Off
Ignition cycle, crash	1,011
ECU life timer, crash	225h:58m:50s
Time from Time zero to End of Event (msec)	109
Time between T-0.5 and Time zero (msec)	215
Lifetime event number	1
Multi-Event Number (Event 1/4)	1

- PRE-CRASH DATA

789/810/825

  - Speed, Vehicle Indicated
  - Accelerator Pedal Position, % full
  - Service Brake (On, Off)
  - ABS Activity (On, Off)
  - Stability Control (On, Off, Engaged)
  - Steering Input
  - Engine RPM
  - PCM Derived Accelerator Pedal Position, % full
  - EV Mode (on, off)
  - Crash Mitigation Braking System, Front Collision Warning System
  - Road Departure Mitigation, Lane Departure Warning System
  - Active Cruise Control
  - Lane Keeping Assist
  - Cruise Control

PRE-CRASH DATA

Gen 4 846 precrash

- Speed, Vehicle Indicated
- PCM derived accelerator pedal position, percent full
- Steering input
- Engine RPM
- Accelerator pedal position, percent full
- Stability control (Engaged/Not Engaged)
- Stability control (On/Off)
- ABS activity (On/Off)
- Service brake (On/Off)
- Collision Mitigation Braking System, Forward Collision Warning (Warning/Not Warning)
- Collision Mitigation Braking System, Forward Collision Warning (Engaged/Not Engaged)
- Collision Mitigation Braking System, Forward Collision Warning (On/Off)
- Lane Keeping Assist (Engaged/Not Engaged)
- Lane Keeping Assist (On/Off)
- Road Departure Mitigation, Lane Departure Warning (Warning/Not Warning)
- Road Departure Mitigation, Lane Departure Warning (Engaged/Not Engaged)
- Road Departure Mitigation, Lane Departure Warning (On/Off)
- Blind Spot information
- Blind Spot information
- Adaptive Cruise Control (Engaged/Not Engaged)
- Adaptive Cruise Control (On/Off)
- Cruise Control (Engaged/Not Engaged)
- Cruise Control (On/Off)
- EV mode (On/Off)

Memo: Reviewed 2024 MY Gen 4 report, no additional changes noted

Honda Gen 2 EDR

- If equipped may have semi autonomous vehicle data elements

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 2 of 3

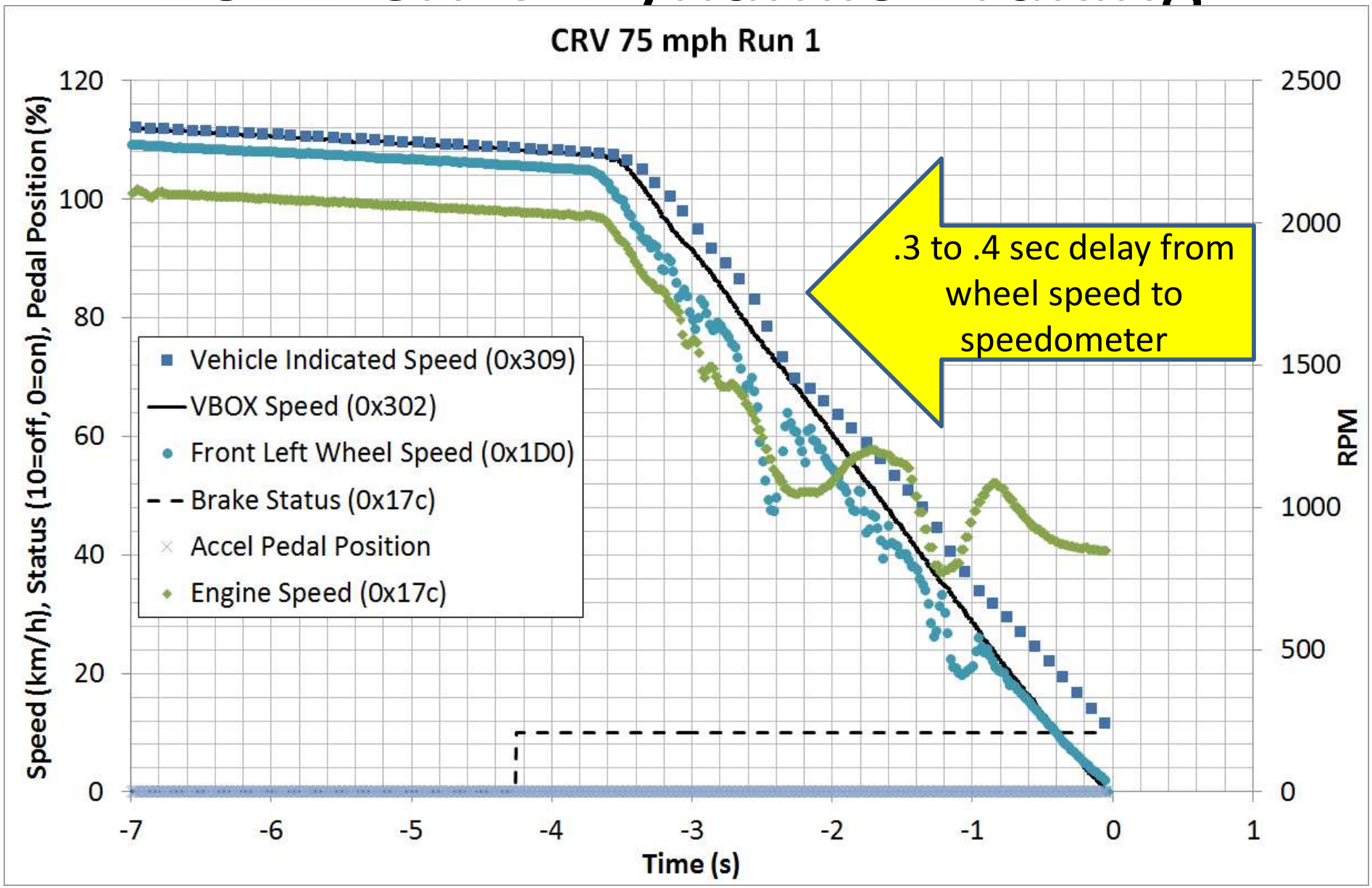
Time Stamp (sec)	PCM Derived Accelerator Pedal Position, % full	Forward Collision Warning (Not Warning/ Warning)	Collision Mitigation Braking System (Not Engaged/ Engaged)	Collision Mitigation Braking System, Forward Collision Warning (On/Off)	Lane Departure Warning (Not Warning/ Warning)	Road Departure Mitigation (Not Engaged/ Engaged)	Road Departure Mitigation, Lane Departure Warning (On/Off)
-5.0	16	Not warning	Not engaged	On	Not warning	Not engaged	Off
-4.5	8	Not warning	Not engaged	On	Not warning	Not engaged	Off
-4.0	11	Not warning	Not engaged	On	Not warning	Not engaged	Off
-3.5	13	Not warning	Not engaged	On	Not warning	Not engaged	Off
-3.0	18	Not warning	Not engaged	On	Not warning	Not engaged	Off
-2.5	18	Not warning	Not engaged	On	Not warning	Not engaged	Off
-2.0	17	Not warning	Not engaged	On	Not warning	Not engaged	Off
-1.5	0	Warning	Engaged	On	Not warning	Not engaged	Off
-1.0	0	Warning	Engaged	On	Not warning	Not engaged	Off
-0.5	0	Warning	Engaged	On	Not warning	Not engaged	Off
0.0	0	Warning	Engaged	On	Not warning	Not engaged	Off

Honda Gen 2 Semi Auton cont'd

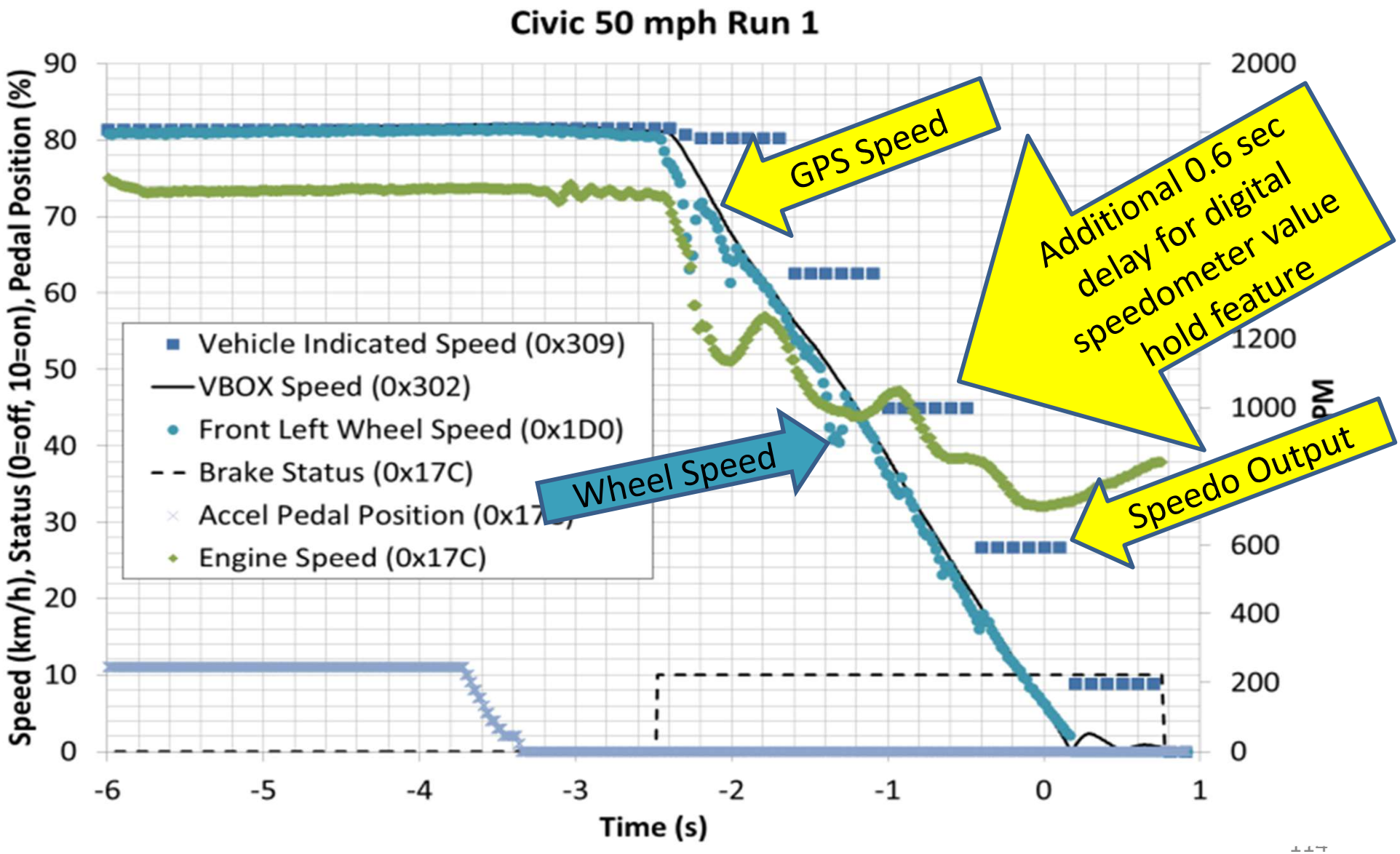
Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 3 of 3

Time Stamp (sec)	Adaptive Cruise Control (Not Engaged/ Engaged)	Adaptive Cruise Control (On/Off)	Lane Keeping Assist (Not Engaged/ Engaged)	Lane Keeping Assist (On/Off)	Cruise Control (Not Engaged/ Engaged)	Cruise Control (On/Off)
-5.0	Not engaged	On	Not engaged	On	Not Engaged	On
-4.5	Not engaged	On	Not engaged	On	Not Engaged	On
-4.0	Not engaged	On	Not engaged	On	Not Engaged	On
-3.5	Not engaged	On	Not engaged	On	Not Engaged	On
-3.0	Not engaged	On	Not engaged	On	Not Engaged	On
-2.5	Not engaged	On	Not engaged	On	Not Engaged	On
-2.0	Not engaged	On	Not engaged	On	Not Engaged	On
-1.5	Not engaged	On	Not engaged	On	Not Engaged	On
-1.0	Not engaged	On	Not engaged	On	Not Engaged	On
-0.5	Not engaged	On	Not engaged	On	Not Engaged	On
0.0	Not engaged	On	Not engaged	On	Not Engaged	On

2012 CR-V Dynamic Braking



2012 Civic Speed Reporting Delay



Honda Summary

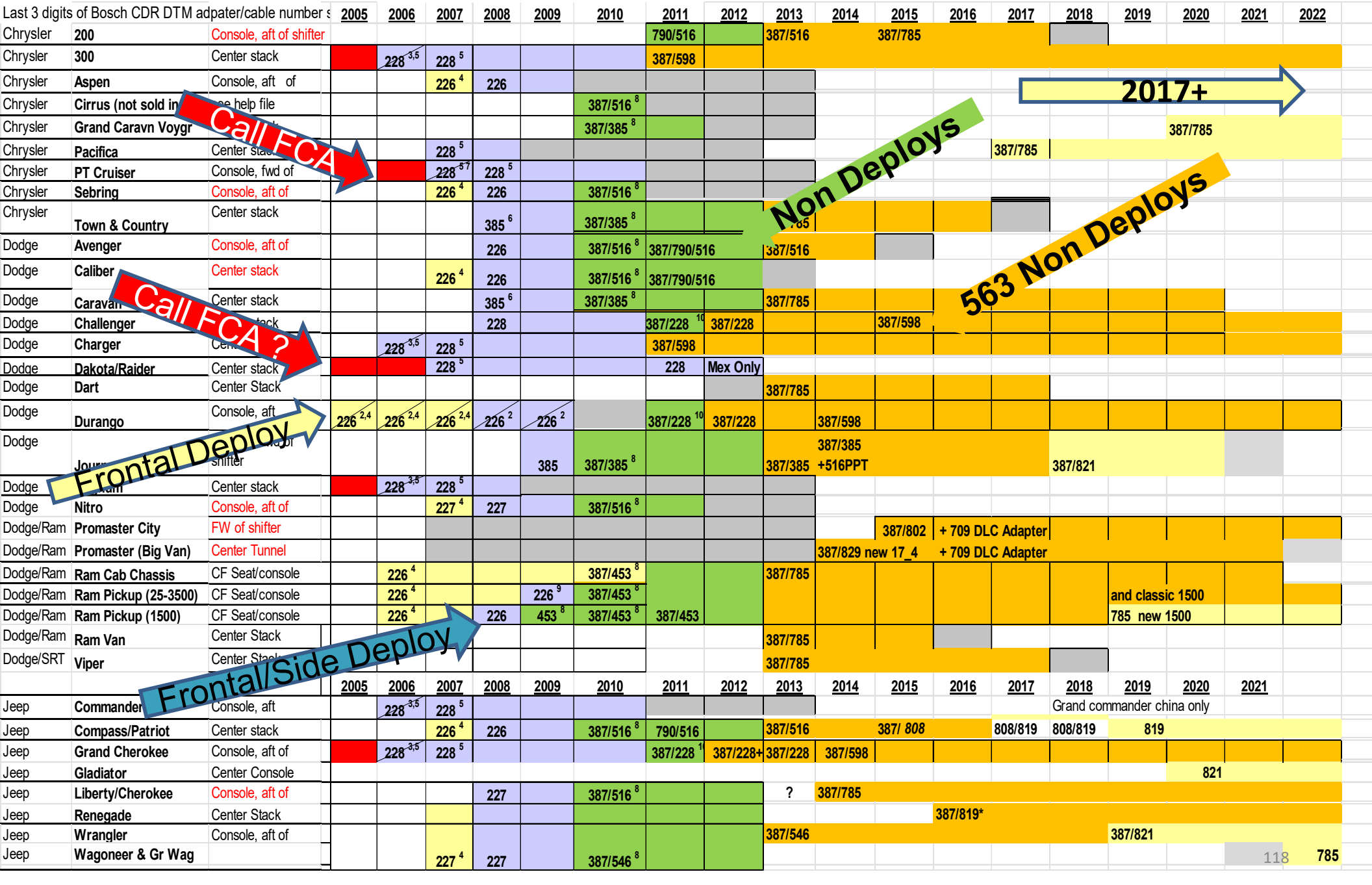
- Honda coverage is of new models only, and begins with 563 intent
- Bonus data includes lateral Delta V, triaxial acceleration, steering, and roll rate.
- Honda records the speed displayed on the speedometer in the EDR, not the speed sent real time from the PCM. Digital speedometers will hold data values for 0.6 seconds. The Honda speed at impact calculation must be adjusted for these delays.

STELLANTIS

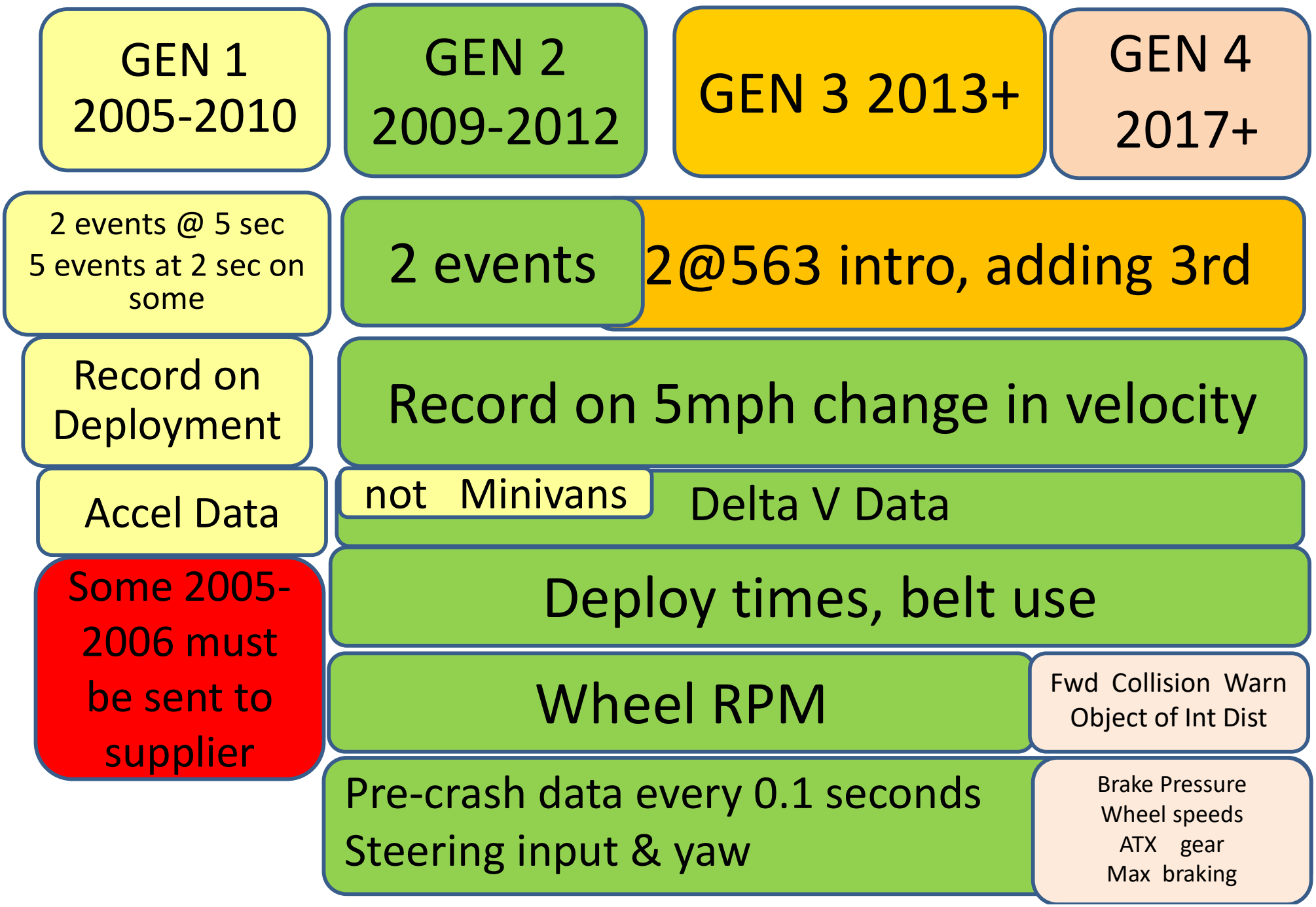
Learning Objectives: Fiat Chrysler

1. *2005 EDR start, data in deployments only*
2. *2009 began phase in of nondeploy events*
3. *Precrash data including stab control and wheel speeds every 0.1 seconds on most*
4. *Thorough data limitations – not model specific*
5. *New Driver Camera module*

Stellantis 21.5 CDR Coverage



FCA EDR Evolution



Chrysler Data Limitations

- One set for all 2005-2012 Models
- Second set for all 2013-current Models
- Read thoroughly, often describes the most common data first then lists exceptions after the general rule.
- Many limitations are for OTHER MODELS than the one you are investigating.
- Due to some non-US production, there are some unusual models

New Jeep/RAM 1500 Driver Camera Module

- Monitors driver during semi autonomous steer assist (hands must remain on wheel, eyes must remain on road)
- Records in deployment or other triggers
- First known instance of a manufacturer making rear (inward) facing camera taking pictures of driver available.
- No DLC imaging, but Grand Cherokee has single on vehicle imaging connector. Others require direct to module and camera attached.
- Imaging instructions in CDR Help File

New Jeep/RAM 1500 Driver Camera Module

- Can store up to 5 events
- Each event can have 5 timestamps in Hex format and 5 images in .Y8 pixel format
  - Y8 often used in machine vision, image processing, and other applications where grayscale needed
- PDF copy does not save pictures, only HEX data
- Images must be exported from .cdrx file
- Irfanview is the recommended viewing format
  - Must download both Installer and All Plugins files
  - Must use exact settings defined in CDR Help File

Driver Camera Module CDR Help File Excerpt

Cable ID# 851 Driver Camera Module

The Driver Monitoring System Module (DMSM) which is part of the Hands-Free Active Driving Assist records driver camera images within the ECU (upon certain triggers such as airbag deployment). The DMSM module records 3 events and can store 5 images for each event. The CDR tool retrieves the recorded image data and stores the data in the CDR file. To export the image data from the CDR file for viewing, select "File>Save Image Data" from the file menu of the CDR application. The Image files are exported and saved in the Crash Data Retrieval folder located in My Documents folder.

Recorded Data

The data recorded includes only retrieved images

Image Data

The data includes recorded images from each event which can be exported by the CDR tool directly into the default data file folder named *Crash Data Retrieval* along with the CDR file. The Images are exported by the CDR tool to their native file format Y = luminance(brightness), U and V means = chrominance (color) with a YUV file extension. During the exporting of the images, the images are named with the user entered VIN, ECU suffix, the record number and the date and time stamp the image was recorded in the ECU.

•<User Entered VIN>_DCM_RecordXImageY_DD-MM-YYY_HH-MM-SS.YUV

The YUV images can be viewed by any off-the-shelf software program that supports the YUV file format such as IrfranView (https://www.irfanview.com/).

See Exporting Images section of the help file for exporting and viewing images

Jeep Driver Camera Module Example PDF File

BOSCH

CDR CRASH DATA RETRIEVAL

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	3C4CJVD15ND******
User	
Case Number	
EDR Data Imaging Date	
Crash Date	
Filename	9145_DCM.CDRX
Saved on	
Imaged with CDR version	Crash Data Retrieval Tool 23.4 rev42b Test Version
Imaged with Software Licensed to (Company Name)	Company Name information was removed when this file was saved without VIN sequence number
Reported with CDR version	Crash Data Retrieval Tool 25.0.817
Reported with Software Licensed to (Company Name)	Dynamic Analysis Group, LLC
EDR Device Type	Drive Camera Module
Event(s) recovered	Record 1, Record 2, Record 3

Comments

No comments entered.

Data Limitations

Driver Camera Module

DATA LIMITATIONS INFORMATION WILL BE UPDATED IN FUTURE RELEASES

The Driver Monitoring System Module (DMSM) which is part of the Hands-Free Active Driving Assist records driver camera images within the ECU (upon certain triggers such as airbag deployment). The DMSM module records 3 events and can store 5 images for each event.

The CDR tool retrieves the recorded image data and stores the data in the CDR file. To export the image data from the CDR file for viewing, select *"File>Save Image Data"* from the file menu of the CDR application. The Image files are exported and saved in the *Crash Data Retrieval* folder located in *My Documents* folder.

Each image file is named with the user entered VIN, ECU suffix, the record number and the date and time stamp the image was recorded in the ECU.

Example image file name: 3C4CJVD15ND96xxxx_DCM_Record1Image1_01-01-2019_00-06-10.YUV

03005_Stellantis_L2_001

BOSCH

CDR CRASH DATA RETRIEVAL

Hexadecimal Data

Data that the vehicle manufacturer has specified for data retrieval is shown in the hexadecimal data section of the CDR report. The hexadecimal data section of the CDR report may contain data that is not translated by the CDR program. The control module contains additional data that is not retrievable by the CDR system.

```
62 29 60 00 01 2D 40 07 01 10 00 00 00 11 00 00
00 10 01 2D 40 11 00 00 07 10 02 5A 80 11 00 00
0E 10 03 87 C0 11 00 00 15 10 04 B5 00 11 00 00
1C

62 29 61 00 01 2D 40 07 01 20 00 00 00 21 00 00
00 20 01 2D 40 21 00 00 07 20 02 5A 80 21 00 00
0E 20 03 87 C0 21 00 00 15 20 04 B5 00 21 00 00
1C

62 29 62 00 01 2D 40 07 01 30 00 00 00 31 00 00
00 30 01 2D 40 31 00 00 07 30 02 5A 80 31 00 00
0E 30 03 87 C0 31 00 00 15 30 04 B5 00 31 00 00
1C

Record 1, Image 1, Time Stamp (DD/MM/YYYY HH:MM:SS) - 01/01/2019 00:06:10
01 01 13 00 06 00 0A

Record 1, Image 2, Time Stamp (DD/MM/YYYY HH:MM:SS) - 01/01/2019 00:06:11
01 01 13 00 06 00 0B

Record 1, Image 3, Time Stamp (DD/MM/YYYY HH:MM:SS) - 01/01/2019 00:06:12
01 01 13 00 06 00 0C

Record 1, Image 4, Time Stamp (DD/MM/YYYY HH:MM:SS) - 01/01/2019 00:06:13
01 01 13 00 06 00 0D

Record 1, Image 5, Time Stamp (DD/MM/YYYY HH:MM:SS) - 01/01/2019 00:06:14
01 01 13 00 06 00 0E

Record 2, Image 1, Time Stamp (DD/MM/YYYY HH:MM:SS) - 02/02/2019 00:00:39
02 02 13 00 00 00 27

Record 2, Image 2, Time Stamp (DD/MM/YYYY HH:MM:SS) - 02/02/2019 00:00:40
02 02 13 00 00 00 28

Record 2, Image 3, Time Stamp (DD/MM/YYYY HH:MM:SS) - 02/02/2019 00:00:41
02 02 13 00 00 00 29

Record 2, Image 4, Time Stamp (DD/MM/YYYY HH:MM:SS) - 02/02/2019 00:00:42
02 02 13 00 00 00 2A

Record 2, Image 5, Time Stamp (DD/MM/YYYY HH:MM:SS) - 02/02/2019 00:00:43
02 02 13 00 00 00 2B

Record 3, Image 1, Time Stamp (DD/MM/YYYY HH:MM:SS) - 03/03/2019 00:00:24
03 03 13 00 00 00 18

Record 3, Image 2, Time Stamp (DD/MM/YYYY HH:MM:SS) - 03/03/2019 00:00:25
03 03 13 00 00 00 19

Record 3, Image 3, Time Stamp (DD/MM/YYYY HH:MM:SS) - 03/03/2019 00:00:26
03 03 13 00 00 00 1A

Record 3, Image 4, Time Stamp (DD/MM/YYYY HH:MM:SS) - 03/03/2019 00:00:27
03 03 13 00 00 00 1B
```